Dimensional Asymptotics of Euler-Based Simulated Likelihood for Multidimensional Diffusions

Diogo Ribeiro
School of Media Arts and Design, Polytechnic of Porto
dfr@esmad.ipp.pt · ORCID 0009-0001-2022-7072

30 July 2026

Abstract

A widely used simulation method for likelihood inference in discretely observed diffusions approximates each transition density by simulating an Euler scheme to one step before the observed endpoint and averaging the final Gaussian density. We show that this endpoint estimator is a shrinking-bandwidth kernel estimator whose effective simulation size is $S/M^{K/2}$, not S, where K is the state dimension, M the number of Euler substeps and S the number of simulations. We derive exact Brownian moments, the general local moment law, the corrected density-level normalisation $\sqrt{S/M^{K/2}}$, and the pointwise mean squared error, which is minimised by the choice $M \asymp S^{2/(K+4)}$. We then show that the simulated density converges in probability to zero, at every pair of points and although the true density is strictly positive, along *every* sequence on the subcritical side of the threshold, that is whenever $S/M^{K/2} \to 0$. The threshold is therefore sharp: the same quantity that must diverge for the central limit theorem to hold must vanish for the estimator to collapse, with no gap between the two conditions. This is not a statement about one badly chosen design: the diagonal $M = S$, which satisfies the rate condition of Brandt and Santa-Clara (2002) and their own regularity assumptions, is merely the corollary that lands inside their hypotheses for every $K > 2$. It refutes the consistency asserted in their Lemma 2 and, since $\sqrt{S}$ times the centred density then diverges, the central-limit statement of their Lemma 3. The proofs of their parameter-estimator theorems consequently fail, and in four dimensions their rate condition is incompatible with mean-square consistency of the simulated density. The threshold is sharp, in that collapse holds throughout the subcritical region and the central limit theorem throughout its complement, and the scaling that produces it follows from two-sided Gaussian bounds on the Euler density rather than from any assumed convergence. **We do not show that the estimator itself is inconsistent**, and in the most tractable case we show the opposite: in a Gaussian location model the simulated maximiser is *consistent* along the very sequences on which the density collapses. The results therefore refute two published statements about the transition density and remove inputs on which the published parameter-estimator proofs rely, while establishing that no argument through pointwise density behaviour alone can settle the maximiser against them. What remains open is the rate, and we set out precisely what a proof would require. Issues specific to the exchange-rate application of that paper, which concern its specification rather than the estimator, are treated in a companion note deposited alongside this one.

Keywords: simulated maximum likelihood; diffusion processes; Euler scheme; transition density; Monte Carlo variance; curse of dimensionality; diffusion bridges; importance sampling.

JEL classifications: C13; C15; C32; G12; G15.

1 Introduction

Continuous-time diffusion models are often observed only at discrete dates. Their exact transition densities are rarely available, which makes exact maximum likelihood difficult even when simulation of approximate sample paths is straightforward. One influential response is the simulation method introduced by Pedersen (1995a) and Pedersen (1995b) and developed and applied by Brandt and Santa-Clara (2002). The method divides each observation interval into M Euler substeps, simulates the process through the first $M-1$ steps, evaluates the Gaussian density of the final Euler step at the observed endpoint, and averages this density over S simulated paths.

For fixed M, the construction is elementary: the sample average converges to the transition density of the M-step Euler chain as $S \to \infty$. For fixed S it is also clear that making the final time step smaller turns the Gaussian endpoint density into an increasingly narrow and increasingly high function. The mathematical question is therefore not whether the two separate approximations are individually meaningful. It is whether they remain compatible when M and S increase jointly.

The starting point is not new and we do not claim it. The final Gaussian density is a kernel with bandwidth $h^{1/2}$, where $h = 1/M$, so the whole construction is a kernel density estimate built from simulated draws. That reading is due to Milstein et al. (2004) and is stated for this estimator by Detemple et al. (2006). In K dimensions the kernel has effective volume of order $h^{K/2}$ and height of order $h^{-K/2}$, so only simulations falling in a shrinking neighbourhood of the endpoint contribute materially, and the effective number of such simulations is of order

$$Sh^{K/2} = \frac{S}{M^{K/2}}.$$

This quantity, rather than S alone, controls the Monte Carlo approximation. That a dimension-dependent rate follows, and that the rate condition of Brandt and Santa-Clara (2002) is inadequate, are also prior results; Section 1.1 sets out exactly what was already known, what this paper adds, and what remains open.

The paper makes six main contributions.

First, we derive exact finite-M moments for the estimator under K-dimensional Brownian motion. The Euler approximation is exact in this benchmark, so any failure is caused by the endpoint Monte Carlo construction and not by discretisation bias, boundary behaviour, nonlinear coefficients, or parameter identification.

Second, we construct a direct counterexample to the joint transition-density convergence asserted in Lemma 2 of Brandt and Santa-Clara (2002). For every $K > 2$, choosing $M = S = n$ makes the simulated density at the Brownian origin converge in probability to zero. Standard Brownian motion satisfies the regularity assumptions of that paper exactly, and the sequence satisfies its condition $\sqrt{S}/M \to 0$, so the counterexample cannot be excluded by strengthening the smoothness, boundedness or ellipticity assumptions alone, since Brownian motion already satisfies all of them in their strongest form. It is excluded only by imposing a joint simulation–discretisation rate condition strong enough to make $S/M^{K/2}$ diverge, which is a restriction on the design rather than on the model. The same sequence also refutes the joint central-limit statement of their Lemma 3, since $\sqrt{S_n}(\widehat{q}_{n,n} - p) \to -\infty$ in probability, so that no finite limit distribution exists. Consequently the published proofs of Theorems 1 and 2 fail, since both rest on those lemmas. In the four-dimensional application, their condition requires $S/M^2 \to 0$, while the variance calculation requires $S/M^2 \to \infty$ for mean-square consistency of the simulated transition density.

We are careful about what this does and does not establish. The counterexample concerns the simulated transition-density estimator at a point. It does not by itself exhibit a sequence along which the simulated-likelihood argmax estimator fails to converge, and we do not claim one. Section 1.3 states the logical status of each result, and Section 7.7 sets out what an argmax counterexample would require.

Third, we establish the generic local moment law for a smooth uniformly elliptic diffusion. If G_h denotes one endpoint-density summand, then for $r > 1$,

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] \longrightarrow (2\pi)^{-(r-1)K/2} r^{-K/2} p(y \mid x)|V(y)|^{-(r-1)/2}.$$

In particular, $\mathrm{Var}(G_h)$ is of order $h^{-K/2}$. This gives the corrected triangular-array central-limit normalisation

$$\sqrt{Sh^{K/2}}\,(\widehat{q}_{M,S} - q_M),$$

not $\sqrt{S}(\widehat{q}_{M,S} - q_M)$.

Fourth, we combine the $O(M^{-1})$ Euler density bias used by Brandt and Santa-Clara (2002) with the corrected simulation variance. The resulting pointwise mean squared error is

$$O(M^{-2}) + O\left(\frac{M^{K/2}}{S}\right),$$

which is minimised at $M \asymp S^{2/(K+4)}$ and yields the rate $S^{-4/(K+4)}$. The estimator therefore exhibits an explicit curse of dimensionality.

Fifth, we trace the density-level result into the log likelihood. The nonlinearity of the logarithm operates on the effective Monte Carlo size $S/M^{K/2}$. Whenever a standard second-order delta expansion is valid, the leading log-density bias is of order $M^{K/2}/S$, rather than uniformly of order $1/S$. This distinction matters because simulated likelihood sums a log transformation over all observations.

Sixth, we evaluate the dimensional law at one implemented design. The exchange-rate application of Brandt and Santa-Clara (2002) uses $N = 544$, $M = 10$, $S = 5{,}000$ in $K = 4$ dimensions, and at that design the ratios corresponding to the paper's own rate conditions are not small, so the implementation does not resemble a point along a sequence in which those conditions operate. We give the arithmetic of what refining M and S together would cost, which is the clearest practical statement of what the effective size $R_{M,S}$ implies. The remaining questions about that application concern its specification rather than its simulation design — square-root boundaries, the reality and invertibility of the incompleteness state, the validity of the Brownian correlation matrix, and the minimum-incompleteness rule — and are treated in the companion note of Section 1.2.

The aim is not to relitigate the substantive economics of a paper published more than two decades ago. Nor do we infer from a false theorem that every numerical estimate obtained with finite M and S must be poor. The aim is narrower and mathematical: to identify the correct dimensional scale of the endpoint estimator, separate sequential from joint convergence, and determine what the published asymptotic arguments do and do not establish.

1.1 Relation to existing asymptotic analyses

The estimator analysed here originates with Pedersen (1995a) and Pedersen (1995b) and is the one implemented by Brandt and Santa-Clara (2002). A substantial literature has already studied it, and it is important to be exact about what that literature establishes, because several statements that might look like the contribution of this paper are older than it.

A note on notation. The cited papers use M, N and δ for quantities that are not the M, N and h of this paper, and silently carrying their symbols across would invert the meaning of several statements. Detemple et al. in particular write M for the number of simulations, which is this paper's S. Throughout this subsection we therefore use neutral symbols: n_{data} for the number of observations, n_{MC} for the number of simulated trajectories, m_{Euler} for the number of Euler substeps, and b_{ker} for a kernel bandwidth, meaning always a standard-deviation scale rather than a variance. In this paper's own notation, $n_{\text{data}} = N$, $n_{\text{MC}} = S$, $m_{\text{Euler}} = M$ and $b_{\text{ker}} = \sqrt{h}$. Footnotes record the original symbol wherever the translation could matter.

What is already known. Milstein et al. (2004, §1, eqs. (1.5)–(1.7)) observe that a simulated transition density of the form $n_{\text{MC}}^{-1} \sum_n \varphi_{b_{\text{ker}}}(\bar{X}_n - y)$, where $\bar{X}_n$ are Euler realisations at the terminal time and $\varphi_{b_{\text{ker}}}$ is a Gaussian kernel, is a special case of a Parzen–Rosenblatt estimator with the bandwidth set equal to the integration step; they then construct forward–reverse representations that are root-n_{MC} consistent and so avoid the resulting curse of dimensionality.[1] Two features of their setting differ from the one studied here, and the difference is where the exponent below comes from.

Their kernel is an *imposed* smoother. In their equation (1.5) it is

$$\varphi_{b_{\text{ker}}}(z) = \left(2\pi b_{\text{ker}}^2\right)^{-K/2} \exp\left(-\|z\|^2/2b_{\text{ker}}^2\right),$$

so its covariance matrix is $b_{\text{ker}}^2 I$ and its bandwidth, in the standard-deviation sense used throughout, is b_{ker}; with b_{ker} set to the integration step this is a covariance of $h^2 I$. It is applied to paths simulated all the way to the terminal time. The endpoint estimator of Pedersen (1995a) and Pedersen (1995b) instead stops one step short and uses the model's own one-step Gaussian density, whose covariance is hV and whose bandwidth is therefore $\sqrt{h}$. Writing both as covariances, theirs is $h^2 I$ and the endpoint kernel's is hV; writing both as bandwidths, theirs is h and the endpoint kernel's is $\sqrt{h}$. Because a K-dimensional kernel of bandwidth b_{ker} has effective volume b_{ker}^K, the effective sample sizes are $n_{\text{MC}} b_{\text{ker}}^K = n_{\text{MC}} h^K$ in their case and $n_{\text{MC}} h^{K/2}$ here. That factor-of-two difference in the exponent is the whole source of the $K/2$ below, and it is a property of which Gaussian is being averaged, not of any choice made here.

The dimension-dependent rate they quote, an optimal bandwidth of order $n_{\text{MC}}^{-1/(4+K)}$ and accuracy $n_{\text{MC}}^{-2/(4+K)}$, is the classical Parzen–Rosenblatt rate for a *free* bandwidth, which they attribute to the nonparametric-statistics literature rather than claiming as their own.

Detemple et al. (2006, p. 32) restate the kernel interpretation for the estimator of Pedersen (1995a) and Pedersen (1995b) and Brandt and Santa-Clara (2002) specifically, and draw three consequences from it. First, the convergence rate is dimension dependent: they give a score rate of order $n_{\text{MC}}^{-2/(K+4)}$, which is the same exponent family as the pointwise rate derived below.[2] Second, optimality requires a joint design linking sample size, simulation size and discretisation level, and they state the conditions explicitly on the same page. Third, and most directly, they observe that Brandt and Santa-Clara (2002) impose one of those two conditions and not the other, and write that this "assumption, unfortunately, is not sufficient" (Detemple et al., 2006, p. 32). They also note, on the same page, that with a fixed discretisation level and a fixed number of simulations the second-order bias explodes as the sample grows, so that asymptotic confidence intervals cover the true value with probability zero.

For a practitioner's map of the alternatives, Hurn et al. (2007) survey and compare the available transition-density methods, including the endpoint construction, on common test problems. Separately, Durham and Gallant (2002) observe that unconditioned Euler paths rarely arrive near the observed endpoint and introduce bridge-based proposals for that reason, and Elerian et al. (2001) augment the path with a Metropolis–Hastings block on the same

[1] They write N for what we call n_{MC} and δ for what we call b_{ker}; their N is a simulation count, not a sample size.

[2] Their notation is $M^{-2/(d+4)}$, where their M is the number of simulations and their d the state dimension; in this paper's symbols that is $S^{-2/(K+4)}$. Their N denotes Euler substeps and their L the number of observations, so all four symbols differ from ours in meaning.

reasoning. Lindström (2012) regularises the Pedersen proposal towards a bridge. Stramer and Yan (2007b) compare the modified-Brownian-bridge simulation approach with closed-form expansions for univariate diffusions, and prove that under constant volatility that sampler is exactly a Brownian bridge; the endpoint estimator appears there only in passing, at their equation (9), as a sampler that "ignores the end point information" (Stramer and Yan, 2007b, p. 677).

Stramer and Yan (2007a) require particular care, because although their theorem concerns a different sampler their remarks about this one are pointed. Their Theorem 1 gives an efficient allocation for the *modified Brownian bridge* sampler, showing that its importance-weight variance is bounded uniformly in m_{Euler} and that the optimal choice is $n_{\mathrm{MC}} \asymp m_{\mathrm{Euler}}^2$ with total error $O(1/m_{\mathrm{Euler}}) + O(1/\sqrt{n_{\mathrm{MC}}})$; they note that this accuracy holds "regardless of the dimension d" (Stramer and Yan, 2007a, Theorem 1, p. 489).[3] About the endpoint sampler they observe three things. It "may result a random variable R_M with unbounded $\mathrm{Var}_q(R_M)$ which leads to inefficiency" (Stramer and Yan, 2007a, p. 486); in its case "a limiting quantity for G_M does not exist and, therefore, even a huge number N can give poor approximations" (Stramer and Yan, 2007a, p. 490); and numerically "for any fixed N, the approximation error of Pedersen's sampler is dominated by the Monte Carlo error and increases as M increases" (Stramer and Yan, 2007a, Fig. 2, p. 494). In the last two quotations their N is again a simulation count. They also record the kernel connection for this estimator specifically, identifying it with a kernel method in which the evaluation points are the preterminal states at time $\Delta - h$ (Stramer and Yan, 2007a, p. 486).

We therefore claim *none* of the following as new: the kernel interpretation of the endpoint density; the dimension dependence of the resulting rate, including the exponent $2/(K+4)$; the existence of a bias–variance trade-off between discretisation and simulation and the associated optimal allocation; the observation that the rate condition of Brandt and Santa-Clara (2002) is inadequate; the need for joint rather than sequential limits; the case for bridge proposals; the possibility that the endpoint estimator's importance-weight variance is unbounded in M; and the observation that increasing S alone need not rescue it.

The same pathology in a different literature. The mechanism identified here is not peculiar to Pedersen's estimator, and saying so places it correctly. In sequential Monte Carlo, Bengtsson et al. (2008) and Snyder et al. (2008) analyse the collapse of importance weights in high dimension for a particle filter whose proposal does not use the observation: the largest weight comes to dominate the sum, the effective sample size falls to one, and avoiding this requires the particle count to grow exponentially in an effective dimension. The endpoint estimator is that construction. Its proposal is the unconditioned Euler path, which ignores the endpoint exactly as an untuned particle filter ignores the observation; its weights are the one-step Gaussian densities; and the shrinking bandwidth $\sqrt{h}$ plays the role of an observation with vanishing error variance, which is the regime in which the weight collapse is sharpest. What Theorem 3.3 adds to that literature is a setting in which the degeneracy is exactly computable, so that an explicit region of collapse can be exhibited rather than a scaling law asserted; what that literature adds here is the assurance that the phenomenon is structural rather than an idiosyncrasy of one estimator. The remedy is also the same on both sides: Beskos, Crisan, et al. (2017) stabilise the high-dimensional filter by localising the resampling, and Rebeschini and Handel (2015) give conditions under which such localisation does beat the dimensional curse; both are the sequential-Monte-Carlo analogue of replacing an unconditioned proposal by one that uses the endpoint.

To restate the point of detail on which the dimensional exponent turns, now in this paper's own symbols: the kernel in the endpoint construction is the model's own one-step Euler density,

[3] They write M for Euler substeps, as we do, but N for the number of simulated trajectories, which is our S; their optimal allocation therefore reads $S \asymp M^2$ in this paper's symbols.

of covariance hV and therefore of bandwidth $\sqrt{h}$, not h. That is why the effective simulation size is $Sh^{K/2}$ rather than Sh^K, and it is the reason a dimension-dependent exponent appears here where Stramer and Yan (2007a) find none for the bridge, whose importance weights carry no shrinking kernel at all.

What this paper adds. Against that background the contributions are the following, each stated for the unconditioned endpoint estimator.

(a) Exact finite-M moments of every order for the endpoint summand under K-dimensional Brownian motion (Proposition 3.1). The prior literature is asymptotic; these are closed forms valid at every M and S, and they are what make the counterexample below computable rather than merely plausible.

(b) The exact local moment constant $c_{r,K} = (2\pi)^{-(r-1)K/2} r^{-K/2}$ for general uniformly elliptic diffusions (Theorem 4.2), together with the identification of $R_{M,S} = S/M^{K/2}$ as the governing quantity. Prior work gives the exponent; this gives the constant.

(c) An explicit collapse-in-probability sequence (Corollary 3.7). The prior negative results are of a different kind: inefficiency, exploding second-order bias in the estimating function, possibly unbounded importance-weight variance, and the warning that a huge N may not help. Each says the estimator may be bad. Corollary 3.7 exhibits a specific sequence, admissible under the published rate condition and the published regularity assumptions, along which the density estimator converges in probability to a value it should not, with the true density strictly positive. That the variance may be unbounded is prior work; that there is an admissible path to the wrong limit is not.

(d) A direct refutation of the joint transition-density statements in Lemmas 2 and 3 of Brandt and Santa-Clara (2002) (Corollary 3.7 and Corollary 3.11), under those authors' own assumptions.

(e) The explicit four-dimensional incompatibility between the published condition, equivalent to $S/M^2 \to 0$, and mean-square consistency, which requires $S/M^2 \to \infty$. Detemple et al. (2006) state that the published condition is insufficient; the contribution here is to show that in the dimension actually used no sequence can satisfy both the published rate condition and mean-square consistency of the simulated transition density. The published condition is not vacuous: many sequences satisfy $\sqrt{S}/M \to 0$, including $S = M$, which is exactly the sequence Corollary 3.7 uses. What fails is their intersection with consistency.

(f) An exact *score* second moment (Proposition 6.4), whose ratio to the density second moment is $(1-h)/\{h(2-h)\}$ and therefore free of K. Differentiating a shrinking-bandwidth kernel estimate costs one further power of the bandwidth, so the score-level effective size is $S/M^{(K+2)/2}$, one power of M more demanding than the density-level $S/M^{K/2}$. In four dimensions this widens the incompatibility of item (e) by a factor of M: the score needs $S/M^3 \to \infty$ where the published condition allows only $S \ll M^2$. Since Lemma 4 of Brandt and Santa-Clara (2002) is a statement about the log likelihood derived from a statement about the density, this is the point at which the published rate condition fails for the object the estimator theorems actually use.

(g) A nondegenerate limit law at the critical boundary (Proposition 3.8): at $K = 2$ along $M = S$ the estimator converges in distribution to a Poisson integral $\sum_i e^{-t_i}$ with mean one and variance one half, completing the trichotomy and showing that the threshold at $K = 2$ separates consistency from collapse without either holding there.

(h) A sharp threshold. Collapse holds on the whole subcritical region, under $R_{M,S} \to 0$ alone (Theorem 3.3), with an explicit rate (Corollary 3.5). The same quantity that must diverge for the density-level central limit theorem must vanish for collapse, so no designs are left unsettled between the two.

(i) The dimensional scaling without the imported convergence hypothesis (Proposition 4.7). Under Aronson-type two-sided Gaussian bounds on the Euler density, which are primitive in μ and V, the moment law holds up to a bracketing of its constant, so the exponent of h and hence the effective size $R_{M,S}$ do not depend on (A5) at all. That assumption is doing exactly one thing: fixing the constant.

(j) A positive answer to the argmax question in the location model. The limit of the nearest-atom criterion (Proposition A.3) and the uniform law that transfers it (Proposition A.4) together give Corollary A.6: the simulated maximiser is consistent along $M = S = n$ with $K > 2$, which is exactly the sequence on which Corollary 3.7 makes the density converge to zero. Collapse of the density and consistency of the maximiser hold simultaneously, so the counterexample provably does not transfer.

(k) The mathematical reassessment of the exchange-rate application, which is reported separately in the companion note described in Section 1.2.

The right description of this paper's relation to Detemple et al. (2006) is therefore sharpening rather than discovery: it derives exact finite-discretisation moments, identifies the local moment constants, and constructs an explicit admissible sequence along which the simulated transition density converges to the wrong value. What distinguishes that last item from an efficiency observation is its logical force. An inefficiency says a design is wasteful and can be repaired by spending more. A collapse along an admissible sequence says the density estimator converges to a value it should not; divergence of S alone does not repair it, and only a fast enough rate of S relative to $M^{K/2}$ does.

Table 1 summarises the comparison.

1.2 The companion note

An earlier version of this paper carried a second half examining the mathematical status of the exchange-rate application of Brandt and Santa-Clara (2002) in detail: the square-root boundaries, the reality of the incompleteness state under the implemented parameterisation, the invertibility of the volatility map, the validity of the Brownian correlation matrix, the minimum-incompleteness identification rule, and the constrained-inference question. That material is now a separate companion note, *Mathematical Status of the Exchange-Rate Application in Brandt and Santa-Clara (2002)*, deposited alongside this paper in the same archive.

The separation is one of kind, not of importance. The results here are asymptotic theorems about an estimator; those there are algebraic propositions and numerical diagnostics about one published specification, and several of them are negative or inconclusive by construction, establishing that no problem arises where one might have been suspected. Nothing in this paper depends on the companion, and the dependence runs one way: the companion cites results from here by number. A reader interested only in the dimensional asymptotics needs nothing from it, and a reader interested only in the application will find the theory summarised there.

One diagnostic from that material is retained here, in Section 8, because it bears directly on the thesis rather than on the application: the implemented design is a concrete point at which the effective size $R_{M,S}$ can be evaluated, and the arithmetic of refining M and S together is the clearest practical statement of what the dimensional law implies.

[4] The entry is "no" in the strict sense used in the column heading, that of exhibiting an admissible sequence along which the estimator converges to an identified wrong value. Stramer and Yan (2007a, p. 490) come closer than a bare "no" conveys: they assert that a limiting quantity for the endpoint construction does not exist, which is a statement of the same kind though not of the same strength.

Table 1: Relation to existing asymptotic analyses of Pedersen-type simulated likelihood. **All entries are translated into this paper's notation**: K state dimension, M Euler substeps, S simulated trajectories, N observations. Several cited papers use these symbols for different quantities, Detemple et al. writing M for the simulation count in particular, so the rates below do not read as in the originals. "Error quantity" names what is converging, since a score rate and a density rate are not comparable. "Joint" asks whether the limits in M, S and N are taken jointly rather than one after another, a sequential limit being weaker. "Counterexample" asks whether a sequence is exhibited along which the estimator converges to a wrong value.

Reference	Estimator	Dim.	Error quantity	Rate or variance result	Joint	Counter-example
Pedersen (1995a) and Pedersen (1995b)	Endpoint	general	estimator	consistency and asymptotic normality	seq.	no
Brandt and Santa-Clara (2002)	Endpoint	applied at $K = 4$	density and estimator	$\sqrt{S}/M \to 0$, $N/S^{1/4} \to 0$	asserted, proved seq.	no
Milstein et al. (2004)	Terminal-time kernel; forward–reverse	explicit in K	density	kernel reading at bandwidth h, covariance $h^2 I$; classical Parzen–Rosenblatt rate; root-S forward–reverse alternative	yes	no
Detemple et al. (2006)	Euler and Doss-transformed, incl. Pedersen and Brandt–Santa-Clara	explicit in K	score	score rate $S^{-2/(K+4)}$; optimal design; published condition "not sufficient"; second-order bias correction	yes	no
Durham and Gallant (2002)	Bridge proposals	general	density, numerically	numerical efficiency comparisons	—	no
Stramer and Yan (2007a)	Modified Brownian bridge; remarks on the endpoint sampler	dimension-free for the bridge	importance weights	bridge weight variance bounded in M, optimal $S \asymp M^2$; endpoint variance may be unbounded and large S may not rescue it	yes	no[4]
Stramer and Yan (2007b)	Modified Brownian bridge; closed-form expansion	univariate	log likelihood, numerically	bridge is exact under constant volatility; comparison run at $S \asymp M^2$	conjectured	no
Lindström (2012)	Regularised bridge	general	density, numerically	sparse-sampling performance	—	no
This paper	Endpoint	explicit in K	density	exact finite-M moments; exact constant $c_{r,K}$; $\mathrm{Var} \asymp M^{K/2}/S$	yes	yes, Corollary 3.7

1.3 Scope of the results

Because this paper both proves theorems and evaluates them at a published design, it matters which claims are which. Every result below falls into one of three categories, and the text signals the category wherever a result is stated. The companion note uses the same scheme, with two further categories it needs and this paper does not: statements reported by the original authors, and implementation details their published description does not settle.

Proved. The exact Brownian moments of Proposition 3.1; the subcritical collapse Theorem 3.3 and its diagonal case Corollary 3.7; the critical limit law Proposition 3.8; the failure of the joint central-limit statement, Corollary 3.11; the general local moment law Theorem 4.2 and its Ornstein–Uhlenbeck instance Proposition 4.4; the uniform bound on the Euler density Lemma 4.11; the density variance scale of Corollary 4.10; the density-level triangular-array central limit theorems Theorem 3.16 and Theorem 4.13; the pointwise mean-squared-error balance of Proposition 5.1; the exact score second moment Proposition 6.4 and the identification Proposition 6.6 that discharges its transfer for constant-coefficient families; the antithetic results Proposition 8.1, Lemma 8.3 and Proposition 8.4; the bracketing of the moment law under Aronson-type bounds Proposition 4.7; and Proposition A.1, Proposition A.2 the nearest-neighbour interchange Proposition A.3, the uniform law Proposition A.4 and the location-model consistency Corollary A.6. The algebraic propositions about the exchange-rate application are stated and proved in the companion note. These are theorems, proved under stated assumptions; those before the semicolon concern the simulated *transition density* or its score, not the parameter estimator.

Conditional. The log-density expansion Proposition 6.1, which requires uniform integrability and lower-tail control that we assume rather than verify; the resulting statements about scores and Hessians; Corollary 6.8, which is proved outright for the constant-coefficient families of Proposition 6.6 and conditional on Assumption 6.5 beyond them; the $\sqrt{N}$ behaviour of the location-model maximiser, whose consistency is proved (Corollary A.6) but whose rate needs the three further ingredients listed in Section A; and the abstract local perturbation result Proposition 6.2, whose hypotheses include consistency of the simulated maximiser and uniform score and Hessian agreement, none of which we verify for the endpoint estimator. It is a conditional abstract result about any pair of maximisers meeting those hypotheses, not an asymptotic theorem for the simulated estimator, and Remark 6.3 states what would be needed to make it one.

Diagnostic. The finite-sample scaling ratios of Section 8 and the numerical evidence of Section 9. These are calculations at particular parameter values and particular designs, not asymptotic theorems, and they are reported as such: they illustrate and check the proved results without extending them. The companion note is diagnostic in the same sense throughout, apart from the algebraic propositions it proves.

One boundary deserves emphasis because it is easy to blur. All the proved results concern the simulated transition density. None of them concerns the maximiser of the simulated log likelihood. We do not prove that the endpoint simulated-likelihood estimator is inconsistent, and Section 7.7 and Section A set out precisely what such a proof would require.

The remainder is organised as follows. Section 2 defines the estimator. Section 3 gives exact Brownian results and the counterexample. Section 4 derives the general diffusion moment law and corrected CLT. Section 5 studies the bias–variance trade-off. Section 6 discusses log likelihood and parameter estimation. Section 7 analyses the published proof. Section 8 gives finite-sample scaling diagnostics for the implemented design. Section 9 presents reproducible numerical evidence, and Section 10 discusses variance-reduction alternatives.

2 The endpoint simulated-likelihood estimator

Let $Y_t \in \mathbb{R}^K$ satisfy the stochastic differential equation

$$dY_t = \mu(Y_t, t; \theta)\, dt + \Sigma(Y_t, t; \theta)\, dW_t, \qquad V = \Sigma\Sigma^\mathsf{T}, \tag{1}$$

where W_t is a Brownian motion of suitable dimension and $\theta \in \mathbf{\Theta} \subset \mathbb{R}^L$. Coefficients are assumed regular enough for a unique strong solution in the sense of Karatzas and Shreve (1991). We suppress t and θ where no ambiguity arises. Consider one observation interval of length one, from $Y_0 = x$ to $Y_1 = y$. The normalisation to unit length is only notational.

For $M \geq 2$, set $h = 1/M$ and define the Euler chain (Kloeden and Platen, 1992)

$$Y_{m+1}^M = Y_m^M + \mu(Y_m^M, mh)h + \Sigma(Y_m^M, mh)\sqrt{h}\, \varepsilon_{m+1}, \qquad m = 0, \ldots, M-1, \tag{2}$$

with $Y_0^M = x$ and independent $\varepsilon_m \sim \mathcal{N}(0, I)$. Conditional on $Y_m^M = z$, the one-step density is

$$g_h(y \mid z) = \varphi_K(y; z + \mu(z, 1-h)h, hV(z, 1-h))\,. \tag{3}$$

Let $Z_h = Y_{M-1}^M$ denote the preterminal Euler state and let $r_h(z \mid x)$ be its density. The M-step Euler transition density is

$$q_M(y \mid x) = \mathbb{E}[g_h(y \mid Z_h)] = \int_{\mathbb{R}^K} g_h(y \mid z) r_h(z \mid x)\, dz. \tag{4}$$

The endpoint simulation estimator is

$$\widehat{q}_{M,S}(y \mid x) = \frac{1}{S} \sum_{s=1}^{S} g_h(y \mid Z_{h,s}), \tag{5}$$

where $Z_{h,1}, \ldots, Z_{h,S}$ are independent draws from the preterminal Euler law. This is equation (8) of Brandt and Santa-Clara (2002), with notation adapted to the present analysis.

For each fixed M, the strong law yields

$$\widehat{q}_{M,S}(y \mid x) \longrightarrow q_M(y \mid x) \quad \text{almost surely as } S \to \infty,$$

provided the summand is integrable. Under regularity conditions such as those of Bally and Talay (1996a) and Bally and Talay (1996b),

$$q_M(y \mid x) - p(y \mid x) = O(M^{-1}),$$

where p is the diffusion transition density. These two facts establish a sequential limit: first $S \to \infty$ for fixed M, then $M \to \infty$. They do not establish convergence along arbitrary joint sequences $M, S \to \infty$.

The geometry of (3) explains why. Its covariance is $hV(z)$. Under nondegeneracy, it is concentrated in a ball of radius $O(\sqrt{h})$ around the endpoint and has maximum height $O(h^{-K/2})$. The integral in (4) remains finite because the effective volume is $O(h^{K/2})$. A Monte Carlo average resolves that integral only if enough preterminal draws enter this shrinking region.

Definition 2.1 (Effective endpoint simulation size)**.** For a K-dimensional endpoint estimator with Euler step $h = 1/M$, define

$$R_{M,S} = Sh^{K/2} = \frac{S}{M^{K/2}}. \tag{6}$$

This is the expected-order number of simulations falling in an $O(\sqrt{h})$ neighbourhood of an endpoint with a regular positive preterminal density.

The next sections make this interpretation exact.

3 Exact Brownian analysis

Consider standard Brownian motion in $\mathbb{R}^K$,

$$dY_t = dW_t, \tag{7}$$

observed over a unit interval. Euler discretisation is exact. This benchmark is not an adversarial construction outside the scope of the published theory: with $\mu \equiv 0$ and $\Sigma \equiv I_K$, all coefficient derivatives vanish and are therefore continuous and bounded of every order, so Assumption 1 of Brandt and Santa-Clara (2002) holds, and $\Sigma\Sigma^\top = I_K$ is positive definite, so Assumption 2 holds. Lemmas 1–3 of that paper are stated under exactly these two assumptions. Any failure exhibited here is therefore a failure inside the stated hypotheses, and cannot be repaired by strengthening the regularity conditions. Set $Y_0 = x$ and evaluate the transition density at $Y_1 = y$. With $h = 1/M$,

$$Z_h \sim \mathcal{N}(x, (1-h)I_K)$$

and a single endpoint summand is

$$G_h = (2\pi h)^{-K/2} \exp\left[-\frac{\|y - Z_h\|^2}{2h}\right]. \tag{8}$$

3.1 Exact moments

Proposition 3.1 (Exact Brownian moments)**.** *Let $r > 0$. For the summand* (8),

$$\begin{aligned}\mathbb{E}[G_h^r] &= (2\pi h)^{-rK/2}\left(1 + \frac{r(1-h)}{h}\right)^{-K/2} \exp\left[-\frac{r\|y-x\|^2}{2\{h + r(1-h)\}}\right] \\ &= (2\pi)^{-rK/2} h^{-(r-1)K/2}\{r - (r-1)h\}^{-K/2} \exp\left[-\frac{r\|y-x\|^2}{2\{r-(r-1)h\}}\right].\end{aligned} \tag{9}$$

In particular,

$$\mathbb{E}[G_h] = (2\pi)^{-K/2} \exp\left[-\frac{\|y-x\|^2}{2}\right] = p(y \mid x), \tag{10}$$

and

$$\mathbb{E}[G_h^2] = (2\pi)^{-K} h^{-K/2}(2-h)^{-K/2} \exp\left[-\frac{\|y-x\|^2}{2-h}\right]. \tag{11}$$

Proof. For $Z \sim \mathcal{N}(x, (1-h)I_K)$ and $c > 0$, the Gaussian identity

$$\mathbb{E}\left[e^{-c\|Z-y\|^2}\right] = \{1 + 2c(1-h)\}^{-K/2} \exp\left[-\frac{c\|x-y\|^2}{1+2c(1-h)}\right]$$

follows by completing the square. Apply it with $c = r/(2h)$ and multiply by the prefactor $(2\pi h)^{-rK/2}$. The cases $r = 1$ and $r = 2$ follow directly. □

Corollary 3.2 (Exact variance)**.** *For S independent simulations,*

$$\mathrm{Var}(\widehat{q}_{M,S}) = \frac{1}{S}\left[(2\pi)^{-K} h^{-K/2}(2-h)^{-K/2} e^{-\|y-x\|^2/(2-h)} - p(y \mid x)^2\right]. \tag{12}$$

Hence

$$\mathrm{Var}(\widehat{q}_{M,S}) \sim \frac{\tau_K^2(x,y)}{Sh^{K/2}}, \qquad \tau_K^2(x,y) = (2\pi)^{-K} 2^{-K/2} e^{-\|y-x\|^2/2}. \tag{13}$$

The variance is not uniformly $O(1/S)$. It is $O(M^{K/2}/S)$. The estimator is unbiased in this example for every M, but its distribution becomes increasingly dominated by rare, very large contributions when $R_{M,S}$ is small.

3.2 A direct joint-limit counterexample

A variance calculation alone does not prove failure of convergence in probability because a non-uniformly integrable sequence can converge in probability while retaining large or divergent moments. The next theorem gives a direct failure result. It is a statement about the simulated transition density at a pair of points; Section 7.7 explains why it does not transfer automatically to the parameter estimator.

The result is stated for the whole subcritical region rather than for one sequence, because the region is what it is about. A reader shown only the diagonal $M = S$ may reasonably answer that one should therefore not set $M = S$; the point is that no path on the subcritical side of the threshold escapes.

Theorem 3.3 (Collapse on the whole subcritical region)**.** *Let $K \geq 1$ and let $x, y \in \mathbb{R}^K$ be arbitrary. Let $h_n = 1/M_n$ with $h_n \leq \frac{1}{2}$, and suppose the design is subcritical in the sense that the effective size of Definition 2.1 vanishes,*

$$R_{M_n,S_n} = S_n h_n^{K/2} \longrightarrow 0. \tag{14}$$

Then the Brownian endpoint estimator satisfies

$$\widehat{q}_{M_n,S_n}(y \mid x) \longrightarrow 0 \qquad \textit{in probability}, \tag{15}$$

although the true density is strictly positive at every such pair,

$$p(y \mid x) = (2\pi)^{-K/2} e^{-\|y-x\|^2/2} > 0.$$

The hypothesis is exactly $R_{M,S} \to 0$, with no logarithmic strengthening. Since $S_n \geq 1$, (14) forces $h_n \to 0$ on its own, so no separate refinement condition is needed. The proof rests on the following exact rewriting, which is worth isolating because it is what makes the effective size visible as the only relevant parameter.

Lemma 3.4 (Exact rescaling)**.** *Write $\lambda_n = S_n h_n^{K/2} = R_{M_n,S_n}$ and let $Z_{n,1}, \ldots, Z_{n,S_n}$ be the preterminal draws, each distributed as $\mathcal{N}(x, (1 - h_n) I_K)$. Then, identically,*

$$\widehat{q}_{M_n,S_n}(y \mid x) = (2\pi)^{-K/2} \frac{T_n}{\lambda_n}, \qquad T_n = \sum_{s=1}^{S_n} \exp\left(-\frac{\|y - Z_{n,s}\|^2}{2h_n} \right). \tag{16}$$

Proof. Each summand of (5) equals $(2\pi h_n)^{-K/2} \exp\{-\|y - Z_{n,s}\|^2/(2h_n)\}$, so

$$\widehat{q}_{M_n,S_n} = S_n^{-1} (2\pi h_n)^{-K/2} T_n = (2\pi)^{-K/2} \frac{T_n}{S_n h_n^{K/2}}. \qquad \square$$

In (16) the estimator is a sum of bounded terms divided by λ_n. Since $\mathbb{E} T_n = (2\pi)^{K/2} \lambda_n p(y \mid x)$, the numerator and the divisor are of the same order in mean, which is unbiasedness; the collapse says that T_n nevertheless falls below any fixed multiple of λ_n with probability tending to one, the mean being carried by a vanishing set.

Proof of Theorem 3.3. Write $\lambda_n = S_n h_n^{K/2}$ and use (16). Fix $R > 0$, to be chosen as a function of n at the end, and split the draws according to whether they lie within $\sqrt{h_n} R$ of y.

Near draws. Let $A_n(R)$ be the event that at least one draw lies in $B(y, \sqrt{h_n} R)$. Since $h_n \leq \frac{1}{2}$ the covariance satisfies $1 - h_n \geq \frac{1}{2}$, so the density of $Z_{n,1}$ is bounded by $\pi^{-K/2}$

everywhere, uniformly in n and in the pair (x,y). With ω_K the volume of the unit ball and $C_K = \pi^{-K/2}\omega_K = 1/\Gamma(\frac{K}{2}+1)$,

$$\mathbb{P}(A_n(R)) \le S_n\,\pi^{-K/2}\omega_K\big(\sqrt{h_n}R\big)^K = C_K\,\lambda_n R^K. \tag{17}$$

This bound does not involve x or y: displacing the evaluation point only moves the ball, and the Gaussian density bound is global.

Far draws. Let T_n^{far} be the part of T_n contributed by draws outside $B(y,\sqrt{h_n}R)$. Substituting $z = y + \sqrt{h_n}u$ and using the same global density bound,

$$\begin{aligned}
\mathbb{E}T_n^{\mathrm{far}} &= S_n \int_{\|z-y\|>\sqrt{h_n}R} \varphi_K(z-x;0,(1-h_n)I_K)e^{-\|z-y\|^2/(2h_n)}\,dz \\
&= S_n h_n^{K/2} \int_{\|u\|>R} \varphi_K(y+\sqrt{h_n}u-x;0,(1-h_n)I_K)e^{-\|u\|^2/2}\,du \\
&\le \lambda_n\,\pi^{-K/2}(2\pi)^{K/2}\,\mathbb{P}(\chi_K > R) = \lambda_n\,2^{K/2}\,\mathbb{P}(\chi_K > R),
\end{aligned} \tag{18}$$

where χ_K denotes the square root of a chi-squared variable with K degrees of freedom. The essential feature of (18) is that the factor multiplying λ_n depends only on R. Markov's inequality therefore gives, for every $\varepsilon > 0$,

$$\mathbb{P}\Big(T_n^{\mathrm{far}} > \varepsilon(2\pi)^{K/2}\lambda_n\Big) \le \frac{2^{K/2}\mathbb{P}(\chi_K > R)}{\varepsilon(2\pi)^{K/2}} = \frac{\mathbb{P}(\chi_K > R)}{\varepsilon\,\pi^{K/2}}, \tag{19}$$

a bound free of λ_n and of n.

Combination. On $A_n(R)^c$ we have $T_n = T_n^{\mathrm{far}}$, so by (16) the event $\{\widehat{q}_{M_n,S_n} > \varepsilon\} \cap A_n(R)^c$ is contained in $\{T_n^{\mathrm{far}} > \varepsilon(2\pi)^{K/2}\lambda_n\}$. Hence, combining (17) and (19),

$$\mathbb{P}(\widehat{q}_{M_n,S_n} > \varepsilon) \le C_K\lambda_n R^K + \frac{\mathbb{P}(\chi_K > R)}{\varepsilon\,\pi^{K/2}} \qquad \text{for every } R > 0. \tag{20}$$

It remains to choose R. Take $R_n = \lambda_n^{-1/(2K)}$, which tends to infinity because $\lambda_n \to 0$. The first term of (20) is then $C_K\lambda_n^{1/2} \to 0$ and the second tends to zero because $R_n \to \infty$. As $\varepsilon > 0$ was arbitrary, (15) follows. $\square$

The two error terms in (20) pull in opposite directions and the theorem needs only that they can be made small *together*. Enlarging R improves the far-field bound and worsens the near-field one; what makes the argument work at $R_{M,S} \to 0$ alone is that the far-field bound (19) does not involve λ_n, so R may be sent to infinity as slowly as required to keep $\lambda_n R^K$ small. Any $R_n \to \infty$ with $\lambda_n R_n^K \to 0$ will do, and such a sequence exists whenever $\lambda_n \to 0$.

Optimising (20) over R rather than taking the convenient choice gives a rate.

Corollary 3.5 (Rate of collapse)**.** *Under the hypotheses of Theorem 3.3, for every $\varepsilon > 0$ there is a constant $C(\varepsilon,K) < \infty$ with*

$$\mathbb{P}(\widehat{q}_{M_n,S_n} > \varepsilon) \le C(\varepsilon,K)\,\lambda_n\{\log(1/\lambda_n)\}^{K/2} \tag{21}$$

for all n with λ_n small enough.

Proof. Take $R_n^2 = 2\log(1/\lambda_n)$ in (20). The first term is $C_K\lambda_n\{2\log(1/\lambda_n)\}^{K/2}$. For the second, the chi tail satisfies $\mathbb{P}(\chi_K > R) \le c_K R^{K-2}e^{-R^2/2}$ for $R \ge 1$, so it is at most $c_K\varepsilon^{-1}\pi^{-K/2}\{2\log(1/\lambda_n)\}^{(K-2)/2}\lambda_n$, which is of smaller order than the first. $\square$

Table 2 evaluates (20) against the exact exceedance probability. The comparison needs no simulation of the full estimator: by (16) each summand depends on its draw only through the distance to y, so the number of draws inside a ball is binomial and the squared radius inside it is a truncated chi-squared, both available in closed form. Designs with $S = 10^7$ are therefore evaluated exactly, and the residual contribution of draws beyond the truncation radius is bounded by 10^{-17} throughout.

Table 2: The bound (20), optimised over R, against the exact probability that the estimator exceeds one tenth of the true density, at $K = 4$ and $x = y = 0$. The designs have $R_{M,S} = 10^{-j}$ at $M = 10^j$, so the effective size falls while the strengthened quantity $R_{M,S}\{\log(1/h)\}^{K/2}$ of the earlier hypothesis is still not small at the top of the table. The bound holds at every design and is tight to within a factor of about four.

M	S	$R_{M,S}$	$R_{M,S}\{\log(1/h)\}^{K/2}$	$\mathbb{P}(\widehat{q} > p/10)$	bound
10	10	10^{-1}	0.530	0.653	6.484
10^2	10^2	10^{-2}	0.212	0.222	1.271
10^3	10^3	10^{-3}	0.0477	0.0375	0.211
10^4	10^4	10^{-4}	0.00848	0.00925	0.0315
10^5	10^5	10^{-5}	0.00133	0.00100	0.00441
10^6	10^6	10^{-6}	0.000191	0.000	0.000588
10^7	10^7	10^{-7}	0.0000260	0.000	0.0000756

The logarithm has not disappeared from the problem; it has moved from the hypothesis to the conclusion, where it belongs. Theorem 3.3 holds on the whole subcritical region, and (21) says the collapse there is fast, essentially linear in the effective size.

Remark 3.6 (What the earlier logarithmic condition was)**.** An earlier form of this theorem assumed $S_n h_n^{K/2}\{\log(1/h_n)\}^{K/2} \to 0$, strictly stronger than (14), and left a region of designs unsettled: those with $R_{M,S} \to 0$ but $R_{M,S}\{\log(1/h)\}^{K/2} \not\to 0$, of which $S_n = h_n^{-K/2}\{\log(1/h_n)\}^{-1}$ with $K > 2$ is an instance. That gap is now closed, and it is worth recording why it was an artefact.

The earlier argument bounded every surviving summand by the largest one and multiplied by S_n, which forces the exclusion radius to satisfy $r_n^2 \asymp h_n \log(1/h_n)$ so that $(2\pi h_n)^{-K/2} e^{-r_n^2/(2h_n)}$ vanishes; the extra logarithm then propagated into the ball-probability bound. Replacing that step by (18), which bounds the far draws by their expectation rather than by S_n times a pointwise maximum, removes the coupling: the far-field bound (19) is a function of R alone. The cost of the crude step was precisely one factor of $\log(1/h_n)$ per dimension.

The diagonal is the case of interest for the published rate condition.

Corollary 3.7 (Collapse along $M = S$)**.** *Let $K > 2$, let $x, y \in \mathbb{R}^K$, and choose $M_n = S_n = n$. Then $\widehat{q}_{n,n}(y \mid x) \to 0$ in probability while $p(y \mid x) > 0$. In every dimension this sequence satisfies the rate condition*

$$\frac{\sqrt{S_n}}{M_n} = n^{-1/2} \longrightarrow 0$$

used in Theorems 1 and 2 of Brandt and Santa-Clara (2002), and it satisfies without any rate restriction the hypotheses of their Lemma 2, which requires only $M \to \infty$ and $S \to \infty$.

Proof. With $h_n = 1/n$ and $S_n = n$ the left side of (14) is $n^{1-K/2}(\log n)^{K/2}$, which tends to zero precisely because $K > 2$. Apply Theorem 3.3. The rate condition is immediate. □

Along $M = S = n$ the effective size is $R_{n,n} = n^{1-K/2}$, so the diagonal is supercritical at $K = 1$, critical at $K = 2$ and subcritical at $K > 2$. The two outer cases are covered by Theorem 3.16 and Corollary 3.7. The critical case is not, and it is worth resolving rather than describing, because at $K = 2$ the estimator neither collapses nor converges: it has a nondegenerate limit law, which can be identified exactly.

Proposition 3.8 (The critical limit law at $K = 2$)**.** *Let $K = 2$, $x = y$, and $M_n = S_n = n$. Then*

$$\frac{\widehat{q}_{n,n}(y \mid x)}{p(y \mid x)} \xrightarrow{d} L := \sum_{i \ge 1} e^{-t_i}, \tag{22}$$

where $\{t_i\}$ are the points of a unit-rate Poisson process on $(0, \infty)$. The limit satisfies $\mathbb{E}L = 1$ and $\operatorname{Var} L = \frac{1}{2}$, and is nondegenerate; consequently $\widehat{q}_{n,n}$ does not *converge in probability to $p(y \mid x)$.*

Proof. With $h = 1/n$ the preterminal draws are $Z_s = x + \sqrt{1-h}\,\xi_s$ with $\xi_s \sim \mathcal{N}(0, I_2)$, and $p(y \mid x) = (2\pi)^{-1}$ at $x = y$. Since $nh = 1$,

$$\frac{\widehat{q}_{n,n}}{p} = \frac{1}{n} \sum_{s=1}^{n} \frac{(2\pi h)^{-1}}{(2\pi)^{-1}} e^{-\|Z_s - y\|^2/(2h)} = \sum_{s=1}^{n} e^{-\|Z_s - y\|^2/(2h)}.$$

In two dimensions $\|\xi_s\|^2 \sim 2E_s$ with E_s independent and standard exponential, so $\|Z_s - y\|^2/(2h) = (1-h)E_s/h = (n-1)E_s$ and

$$\frac{\widehat{q}_{n,n}}{p} = \sum_{s=1}^{n} e^{-(n-1)E_s}, \tag{23}$$

an identity, not an approximation. The points $\{(n-1)E_s\}_{s \le n}$ form n independent draws from the exponential law of rate $1/(n-1)$; for any $0 \le a < b$ the expected count in $[a, b]$ is $n\{e^{-a/(n-1)} - e^{-b/(n-1)}\} \to b - a$, and the counts in disjoint intervals are asymptotically independent, so the point process converges to a unit-rate Poisson process Π on $(0, \infty)$. Since $t \mapsto e^{-t}$ is bounded, continuous and integrable, the mapping $\Pi \mapsto \int e^{-t}\Pi(dt)$ is continuous for the vague topology on the relevant configurations, and (22) follows.

The moments are exact at every n and identify the limit's. From (23), $\mathbb{E}e^{-(n-1)E} = 1/n$, so $\mathbb{E}[\widehat{q}_{n,n}/p] = 1$ for every n, which is Proposition 3.1 at $r = 1$; and $\mathbb{E}e^{-2(n-1)E} = 1/(2n-1)$ gives

$$\operatorname{Var}\Big(\frac{\widehat{q}_{n,n}}{p}\Big) = \frac{n}{2n-1} - \frac{1}{n} \longrightarrow \frac{1}{2},$$

which agrees with Proposition 3.1 at $r = 2$. For the limit, $\mathbb{E}L = \int_0^\infty e^{-t}dt = 1$ and $\operatorname{Var} L = \int_0^\infty e^{-2t}dt = \frac{1}{2}$ by the Campbell formulas. A random variable with positive variance is not a constant, so $L \neq p/p = 1$ almost surely fails, and convergence in distribution to a nondegenerate limit precludes convergence in probability to the constant 1. $\square$

Remark 3.9 (What the critical case shows, and what it repairs)**.** Three points. First, the trichotomy along the diagonal is now complete: consistency and asymptotic normality at $K = 1$, a nondegenerate Poisson-integral limit at $K = 2$, collapse at $K > 2$. Dimension two is a genuine boundary rather than an artefact of the proof technique.

Second, the argument for inconsistency at $K = 2$ is a limit law and not a variance calculation. That distinction matters here more than usual, because Section 3 opens by warning that a non-uniformly integrable sequence can converge in probability while retaining large or even divergent moments; inferring inconsistency at $K = 2$ from $\operatorname{Var} \to \frac{1}{2}$ alone would be exactly that fallacy. The two computations do agree, and the agreement is a useful check, but it is (22) that carries the conclusion.

Third, the limit exhibits the mechanism in its simplest form. The estimator is a sum over a Poisson process of exponentially decaying contributions, so it is dominated by the few draws that land closest to the endpoint; its mean is one because the Campbell integral is one, while its median is about 0.89 and it falls below half the true density with probability about 0.28. That is the same rare-event geometry that drives Corollary 3.7, held at the boundary where it neither vanishes nor explodes, and it is what the qualitative discussion of Remark 3.13 describes.

Remark 3.10 (The collapse is not an artefact of the evaluation point)**.** Neither bound in the proof depends on x or y: the first is a global Gaussian density bound applied to a ball centred at y, and the second uses only that every draw lies outside that ball. The collapse is therefore not a feature of evaluating at the mode, where one might suspect the estimator is stressed unusually, nor of the coincident case $x = y$. It holds at every pair of points, and the limit is zero at all of them while $p(y \mid x)$ is strictly positive at all of them. What varies with (x, y) is only the constant in Proposition 3.1, not the rate or the conclusion.

The same sequence refutes a second published statement. Lemma 3 of Brandt and Santa-Clara (2002) asserts that, under $\sqrt{S}/M \to 0$,

$$\sqrt{S}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\}$$

converges in distribution to a normal law. That cannot hold along $M_n = S_n = n$.

Corollary 3.11 (Failure of the published joint central-limit statement)**.** *Let $K > 2$ and $x = y = 0$, and take $M_n = S_n = n$, a sequence satisfying $\sqrt{S_n}/M_n \to 0$. Then*

$$\sqrt{S_n}\,\{\widehat{q}_{n,n}(0 \mid 0) - p(0 \mid 0)\} \longrightarrow -\infty \qquad \textit{in probability,} \tag{24}$$

so the sequence converges in distribution to no finite limit, normal or otherwise. The joint central-limit statement of Lemma 3 is therefore false along an admissible sequence, whatever variance is assigned to the limit.

Proof. Write $p = p(0 \mid 0) = (2\pi)^{-K/2} > 0$ and fix $\Lambda > 0$. By Corollary 3.7, $\widehat{q}_{n,n} \to 0$ in probability, so with $\varepsilon = p/2$,

$$\mathbb{P}\left(\widehat{q}_{n,n} > \tfrac{p}{2}\right) \longrightarrow 0.$$

On the complementary event, $\widehat{q}_{n,n} - p \le -\frac{p}{2}$, hence $\sqrt{S_n}(\widehat{q}_{n,n} - p) \le -\frac{p}{2}\sqrt{n}$. Since $\frac{p}{2}\sqrt{n} > \Lambda$ for all n large,

$$\mathbb{P}\left(\sqrt{S_n}\,\{\widehat{q}_{n,n} - p\} > -\Lambda\right) \le \mathbb{P}\left(\widehat{q}_{n,n} > \tfrac{p}{2}\right) \longrightarrow 0,$$

which is (24). A sequence diverging to $-\infty$ in probability is not tight, so it has no weak limit in $\mathbb{R}$. $\square$

Remark 3.12 (What is and is not refuted)**.** It is worth stating the hierarchy in one place, since the four statements are easy to run together.

(i) The joint transition-density consistency of Lemma 2 is *directly false* (Corollary 3.7).

(ii) The joint central-limit statement of Lemma 3 is *directly false* along the same sequence (Corollary 3.11). Note also that the limiting variance it states is the variance of a summand that itself depends on M, so the statement is ill-posed as an asymptotic claim even before the counterexample is applied.

(iii) The published proofs of Theorems 1 and 2 *fail*, since Lemmas 4 to 6 take Lemmas 2 and 3 as inputs.

(iv) The conclusions of the parameter-estimator theorems are *not* disproved. The construction says nothing directly about the simulated-likelihood argmax; see Section 7.7 and Section A.

Remark 3.13 (How an unbiased estimator converges to zero)**.** For every n, $\mathbb{E}[\widehat{q}_{n,n}] = p(0 \mid 0)$. The convergence in probability to zero is sustained by increasingly rare simulations with increasingly large endpoint densities. The sequence is not uniformly integrable. This is precisely the rare-event geometry hidden by a fixed-M application of the strong law.

Corollary 3.14 (Contradiction in four dimensions)**.** *For $K = 4$, mean-square consistency requires*

$$\frac{S}{M^2} \longrightarrow \infty,$$

whereas the published condition $\sqrt{S}/M \to 0$ is equivalent to

$$\frac{S}{M^2} \longrightarrow 0.$$

The two conditions are mutually exclusive. Moreover, Corollary 3.7 shows that the published condition permits a sequence along which the density estimator converges to the wrong value.

Remark 3.15 (Where the conflict is unconditional)**.** Two words are worth separating before this is read. A sequence is *admissible* when it satisfies the published assumptions, in particular $\sqrt{S}/M \to 0$; it is *density-consistent* when $S/M^{K/2} \to \infty$, which is what mean-square consistency of the simulated transition density requires. These are different properties, and the point below is that in four dimensions no sequence has both, not that either class is empty. Admissible sequences abound, $S = M$ among them.

The incompatibility in Corollary 3.14 is a property of dimensions $K \geq 4$. Mean-square consistency requires $S/M^{K/2} \to \infty$ and the published condition requires $S/M^2 \to 0$; for $K \geq 4$ the first demands $S \gtrsim M^{K/2} \geq M^2$ and the second demands $S \ll M^2$, so no sequence satisfies both. For $K \leq 3$ the two are compatible: with $K = 3$ and $S = M^{7/4}$ one has $S/M^{3/2} \to \infty$ and $\sqrt{S}/M \to 0$ simultaneously. Corollary 3.7 is nevertheless not confined to $K \geq 4$. It shows that even when the published condition admits consistent sequences, as at $K = 3$, it also admits inconsistent ones, so the condition cannot be what makes the theorem work. The empirical application of Brandt and Santa-Clara (2002) has $K = 4$, where no admissible sequence is also density-consistent.

3.3 Correct triangular-array central limit theorem

Theorem 3.16 (Brownian endpoint CLT)**.** *Let $M \to \infty$ and $S \to \infty$ such that*

$$Sh^{K/2} = \frac{S}{M^{K/2}} \longrightarrow \infty. \tag{25}$$

Then

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\} \Longrightarrow \mathcal{N}\Big(0, \tau_K^2(x,y)\Big), \tag{26}$$

where $\tau_K^2(x,y)$ is defined in (13)*.*

Proof. Index the array by n, with $h_n = 1/M_n$ and $s = 1, \ldots, S_n$; within row n the summands $G_{n,s}$ are independent and identically distributed. Proposition 3.1 at $r = 2$ gives $h_n^{K/2}\mathbb{E}[G_{n,1}^2] \to \tau_K^2$, and $\mathbb{E}G_{n,1} = p(y \mid x)$ exactly for every M, so

$$h_n^{K/2}\,\mathrm{Var}(G_{n,1}) = h_n^{K/2}\mathbb{E}[G_{n,1}^2] - h_n^{K/2}p(y \mid x)^2 \longrightarrow \tau_K^2.$$

At $r = 3$ the same proposition gives $\mathbb{E}[G_{n,1}^3] = O(h_n^{-K})$, a bound on the *raw* moment. Since $G_{n,1} \geq 0$ and $\mathbb{E}G_{n,1} = p(y \mid x)$ is constant in n, the inequality $|a - b|^3 \leq 4(a^3 + b^3)$ for $a, b \geq 0$ gives the centred bound

$$\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3 \leq 4\left\{\mathbb{E}[G_{n,1}^3] + p(y \mid x)^3\right\} = O(h_n^{-K}).$$

The Lyapunov ratio at order three is therefore

$$\frac{S_n \mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3}{\{S_n \operatorname{Var}(G_{n,1})\}^{3/2}} = \frac{S_n O(h_n^{-K})}{S_n^{3/2} \tau_K^3 h_n^{-3K/4}\{1 + o(1)\}} = O\left(\left(S_n h_n^{K/2}\right)^{-1/2}\right) \longrightarrow 0$$

by (25). The triangular-array Lyapunov central limit theorem applies. Because the Euler scheme is exact for Brownian motion, $q_M = p$ for every M and no bias term appears, so the statement may be centred at p directly. ☐

The correct normalisation depends on dimension. A fixed-M $\sqrt{S}$ central limit theorem is valid for each M separately, but it is not stable as $M \to \infty$: see Remark 4.14.

4 General multidimensional diffusions

The Brownian calculation reflects a local property of Gaussian endpoint kernels. We now derive the generic moment law. We state it under a clean global framework, so that every step of the proof is checkable and no condition is phrased in terms of the conclusion it is meant to deliver; Remark 4.5 explains how localisation would weaken it.

Throughout this section the model is time homogeneous, so that $\mu(z)$ and $V(z)$ carry no time argument. The one-step kernel (3) is then evaluated at the preterminal state alone, and no ambiguity arises about whether coefficients are taken at time $1-h$ or at time 1. Remark 4.9 records what changes otherwise.

Let $G_h = g_h(y \mid Z_h)$ be one summand in (5), where Z_h is the Euler chain at time $1 - h$. Fix x and y.

Assumption 4.1 (Global regularity)**.** *(A1) $\mu : \mathbb{R}^K \to \mathbb{R}^K$ is bounded and continuous, with $\|\mu\|_\infty = \bar{\mu} < \infty$.*

(A2) $V : \mathbb{R}^K \to \mathbb{R}^{K \times K}$ is continuous and globally uniformly elliptic: there exist constants $0 < \underline{v} \leq \overline{v} < \infty$ with

$$\underline{v}\, I_K \preceq V(z) \preceq \overline{v}\, I_K \qquad \text{for every } z \in \mathbb{R}^K. \tag{27}$$

(A3) For each $h \in (0, \frac{1}{2}]$ the preterminal Euler state Z_h, that is the chain (2) at time $1 - h$, has a density $r_h(\cdot \mid x)$ with respect to Lebesgue measure.

(A4) The family is uniformly bounded: $\sup_{0<h\leq 1/2} \sup_{z \in \mathbb{R}^K} r_h(z \mid x) = \bar{r} < \infty$.

(A5) There is a continuous density $p(\cdot \mid x)$ with $p(y \mid x) > 0$ such that $r_h(\cdot \mid x) \to p(\cdot \mid x)$ uniformly on some closed ball $\overline{B}(y, \delta_0)$, $\delta_0 > 0$, as $h \downarrow 0$.

Two comments on the content of these conditions. (A5) is a joint requirement in time and space: the preterminal state lives at time $1 - h$, not at time 1, so the stated convergence combines convergence of the Euler law to the diffusion law with continuity of $t \mapsto p_t(\cdot \mid x)$ at $t = 1$ in the topology of uniform convergence near y. The limit $p(\cdot \mid x)$ is the diffusion transition density at time 1. And (A4) is not implied by (A3): it is a uniform-in-h statement, which for uniformly elliptic coefficients follows from the Gaussian upper bounds for Euler densities established by Bally and Talay (1996b), since the preterminal time $1 - h \geq \frac{1}{2}$ is bounded away from zero.

Assumption 4.1 is compatible with the regular setting used to justify Euler density expansions in Bally and Talay (1996a) and Bally and Talay (1996b). It is not compatible with the square-root boundaries of the exchange-rate model examined in the companion note, where (A2) fails at the origin; that distinction is discussed there.

The following theorem proves the dimensional moment law under Assumption 4.1. It concerns the simulated transition density.

Theorem 4.2 (Local moment law). *Under Assumption 4.1, for every fixed $r \geq 1$,*

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] \longrightarrow c_{r,K}\, p(y \mid x)\, |V(y)|^{-(r-1)/2}, \tag{28}$$

where

$$c_{r,K} = (2\pi)^{-(r-1)K/2} r^{-K/2}. \tag{29}$$

In particular,

$$h^{K/2}\mathbb{E}[G_h^2] \longrightarrow (4\pi)^{-K/2} p(y \mid x)|V(y)|^{-1/2}. \tag{30}$$

Proof. Write $m_h(z) = y - z - h\mu(z)$, so that by (3)

$$g_h(y \mid z)^r = (2\pi h)^{-rK/2}|V(z)|^{-r/2} \exp\left[-\frac{r}{2h} m_h(z)^\mathsf{T} V(z)^{-1} m_h(z)\right], \tag{31}$$

and fix $\delta \in (0, \delta_0]$. Split

$$\begin{aligned} h^{(r-1)K/2}\mathbb{E}[G_h^r] &= I_h(\delta) + T_h(\delta), \\ I_h(\delta) &= h^{(r-1)K/2} \int_{B(y,\delta)} g_h(y \mid z)^r\, r_h(z \mid x)\, dz, \\ T_h(\delta) &= h^{(r-1)K/2} \int_{B(y,\delta)^c} g_h(y \mid z)^r\, r_h(z \mid x)\, dz. \end{aligned}$$

Tail term. For $z \notin B(y, \delta)$ and $h \leq \delta/(4\bar{\mu} + 1)$, (A1) gives

$$\|m_h(z)\| \geq \|y - z\| - h\bar{\mu} \geq \tfrac{1}{2}\|y - z\| \qquad \text{and} \qquad \|m_h(z)\| \geq \tfrac{1}{2}\delta.$$

By (A2) we have $V(z)^{-1} \succeq \overline{v}^{-1} I_K$ and $|V(z)|^{-r/2} \leq \underline{v}^{-rK/2}$, so

$$g_h(y \mid z)^r \leq (2\pi h)^{-rK/2} \underline{v}^{-rK/2} \exp\left[-\frac{r\|m_h(z)\|^2}{2\overline{v}h}\right].$$

Splitting the exponent in half and using the two lower bounds separately,

$$g_h(y \mid z)^r \leq (2\pi h)^{-rK/2} \underline{v}^{-rK/2} \exp\left[-\frac{r\delta^2}{16\overline{v}h}\right] \exp\left[-\frac{r\|y - z\|^2}{16\overline{v}h}\right].$$

Using (A4) and $\int_{\mathbb{R}^K} e^{-r\|w\|^2/(16\overline{v}h)} dw = (16\pi\overline{v}h/r)^{K/2}$, and collecting powers as $h^{(r-1)K/2} \cdot h^{-rK/2} \cdot h^{K/2} = 1$,

$$T_h(\delta) \leq \bar{r}\, \underline{v}^{-rK/2} (2\pi)^{-rK/2} (16\pi\overline{v}/r)^{K/2} \exp\left[-\frac{r\delta^2}{16\overline{v}h}\right] \xrightarrow[h\downarrow 0]{} 0. \tag{32}$$

Local term. Substitute $z = z_h(u) = y + \sqrt{h}\, u$, so that $dz = h^{K/2} du$ and the domain becomes $\|u\| < \delta/\sqrt{h}$. Since $h^{(r-1)K/2}(2\pi h)^{-rK/2} h^{K/2} = (2\pi)^{-rK/2}$, every power of h cancels and

$$I_h(\delta) = (2\pi)^{-rK/2} \int_{\mathbb{R}^K} \mathbf{1}\{\|u\| < \delta/\sqrt{h}\}\, \Psi_h(u)\, du,$$

$$\Psi_h(u) = |V(z_h(u))|^{-r/2} r_h(z_h(u) \mid x) \exp\Big[-\frac{r}{2h} m_h(z_h(u))^\mathsf{T} V(z_h(u))^{-1} m_h(z_h(u))\Big].$$

Here $m_h(z_h(u)) = -\sqrt{h}\, u - h\mu(z_h(u))$, so

$$\frac{1}{h} m_h(z_h(u))^\mathsf{T} V(z_h(u))^{-1} m_h(z_h(u)) = (u + \sqrt{h}\,\mu(z_h(u)))^\mathsf{T} V(z_h(u))^{-1}(u + \sqrt{h}\,\mu(z_h(u))).$$

Fix u. Then $z_h(u) \to y$, so continuity of μ and V gives $V(z_h(u)) \to V(y)$ and $\sqrt{h}\,\mu(z_h(u)) \to 0$, and the quadratic form converges to $u^\mathsf{T} V(y)^{-1} u$. For h small enough that $\sqrt{h}\|u\| \le \delta_0$, (A5) and continuity of $p(\cdot \mid x)$ give $r_h(z_h(u) \mid x) \to p(y \mid x)$. Hence, pointwise in u,

$$\Psi_h(u) \longrightarrow |V(y)|^{-r/2} p(y \mid x) \exp\Big[-\tfrac{r}{2} u^\mathsf{T} V(y)^{-1} u\Big].$$

Domination. Let $h \le \min\{\delta_0, (2\bar\mu)^{-2}\}$. If $\|u\| \ge 2\sqrt{h}\,\bar\mu$ then $\|u + \sqrt{h}\,\mu(z_h(u))\| \ge \frac{1}{2}\|u\|$, so by (A2) and (A4)

$$\Psi_h(u) \le \underline{v}^{-rK/2}\, \bar r\, \exp\Big[-\frac{r\|u\|^2}{8\bar v}\Big],$$

while if $\|u\| < 2\sqrt{h}\,\bar\mu \le 1$ then $\Psi_h(u) \le \underline{v}^{-rK/2}\bar r$. The dominating function

$$\Lambda(u) = \underline{v}^{-rK/2}\bar r \left\{\exp\Big[-\frac{r\|u\|^2}{8\bar v}\Big] + \mathbf{1}\{\|u\| \le 1\}\right\}$$

is integrable and does not depend on h. Dominated convergence therefore gives

$$\lim_{h\downarrow 0} I_h(\delta) = (2\pi)^{-rK/2} |V(y)|^{-r/2} p(y \mid x) \int_{\mathbb{R}^K} \exp\Big[-\tfrac{r}{2} u^\mathsf{T} V(y)^{-1} u\Big]\, du.$$

The Gaussian integral equals $(2\pi/r)^{K/2}|V(y)|^{1/2}$, so

$$\lim_{h\downarrow 0} I_h(\delta) = (2\pi)^{-rK/2}(2\pi/r)^{K/2} p(y \mid x)|V(y)|^{-(r-1)/2} = c_{r,K}\, p(y \mid x)|V(y)|^{-(r-1)/2}.$$

This limit does not depend on δ, and $T_h(\delta) \to 0$ by (32), which proves (28). Taking $r = 2$ and using $(2\pi)^{-K/2}2^{-K/2} = (4\pi)^{-K/2}$ gives (30). □

Remark 4.3 (Verification in exactly solvable cases)**.** Two remarks on the statement before the constant is checked. First, $r = 1$ is admissible and the proof covers it verbatim; there $c_{1,K} = 1$ and the conclusion reads $\mathbb{E}[G_h] = q_M(y \mid x) \to p(y \mid x)$, so the convergence of the Euler density is the $r = 1$ case of the moment law rather than a separate input. Lemma 4.11 remains useful because it supplies a bound uniform in h with an explicit constant, which is what the central limit theorem needs, and it does so without invoking (A5).

Second, the conclusion holds well outside Assumption 4.1, and one case can be settled completely rather than asserted. The Ornstein–Uhlenbeck process has unbounded drift and so violates (A1); it is also the process used for the numerical check in Section 9. For it the preterminal law is exactly Gaussian and the moment law can be proved directly.

Proposition 4.4 (The moment law for Ornstein–Uhlenbeck processes)**.** *Let $dY_t = -\alpha Y_t\, dt + \sigma\, dW_t$ in $\mathbb{R}^K$ with $\alpha > 0$ and $\sigma > 0$, so that $V \equiv \sigma^2 I_K$. Then for every $x, y \in \mathbb{R}^K$ and every $r \ge 1$, the conclusion* (28) *of Theorem 4.2 holds, with the same constant $c_{r,K} = (2\pi)^{-(r-1)K/2} r^{-K/2}$.*

Proof. The Euler chain at time $1 - h$ has the exact Gaussian law $Z_h \sim \mathcal{N}(m_h, v_h I_K)$ for some $m_h \to xe^{-\alpha}$ and $v_h \to v_1 = \sigma^2(1 - e^{-2\alpha})/(2\alpha) > 0$, and its density r_h converges to the diffusion density $p(\cdot \mid x)$ uniformly on compacts, this last because both are Gaussian with $m_h \to xe^{-\alpha}$ and $v_h \to v_1 > 0$, so the densities converge pointwise and, being uniformly bounded with uniformly bounded derivatives on a compact set, uniformly there. The summand is $G_h = \varphi_{h\sigma^2 I}(y - (1-\alpha h)Z_h)$. Substituting $u = y - (1-\alpha h)z$, whose Jacobian is $(1-\alpha h)^{-K}$,

$$\mathbb{E}[G_h^r] = (1-\alpha h)^{-K} \int \varphi_{h\sigma^2 I}(u)^r \, r_h\Big(\frac{y-u}{1-\alpha h}\Big) du.$$

Now $\varphi_{h\sigma^2 I}(u)^r = (2\pi h\sigma^2)^{-(r-1)K/2} r^{-K/2} \varphi_{(h\sigma^2/r)I}(u)$, an exact identity, so

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] = (1-\alpha h)^{-K}(2\pi\sigma^2)^{-(r-1)K/2} r^{-K/2}\, \mathbb{E}\Big[r_h\Big(\tfrac{y-U_h}{1-\alpha h}\Big)\Big], \qquad U_h \sim \mathcal{N}(0, \tfrac{h\sigma^2}{r} I).$$

As $h \downarrow 0$ the first factor tends to one and $U_h \to 0$ in probability, so the expectation tends to $p(y \mid x)$ by uniform convergence of r_h on a neighbourhood of y together with continuity of p. Since $|V|^{-(r-1)/2} = \sigma^{-K(r-1)} = (\sigma^2)^{-(r-1)K/2}$, the limit is $c_{r,K}\, p(y \mid x)\, |V(y)|^{-(r-1)/2}$. □

This is what Figure 4 illustrates, and it means the figure now confirms a proposition rather than gesturing beyond a theorem's hypotheses. It also indicates what the general localised version would need: not global bounds on μ and V, but ellipticity and drift boundedness near y together with a Gaussian upper bound on r_h of Aronson type, which is what Bally and Talay (1996a) and Bally and Talay (1996b) and Konakov and Mammen (2002) establish for Euler schemes and which would subsume (A4) and the tail control at once. We state Assumption 4.1 globally because it keeps the proof short, and record the localisation as the natural strengthening rather than carrying it out.

The constant can be checked against closed forms. For standard Brownian motion $V \equiv I_K$, and substituting $p(y \mid x) = (2\pi)^{-K/2} e^{-\|y-x\|^2/2}$ into (28) reproduces the exact limit of Proposition 3.1 term by term. For a constant-coefficient diffusion $dY = \mu\, dt + \Sigma\, dW$ the same computation with $V = \Sigma\Sigma^\top$ reproduces (28) including the determinant factor. For a multidimensional Ornstein–Uhlenbeck process the preterminal law is Gaussian and the second moment is again available in closed form; Figure 4 confirms convergence to (30) numerically. The cases $r = 2$ and $r = 3$ supply exactly the second and third moments used in Theorem 4.13.

Remark 4.5 (Localisation)**.** Only three features of Assumption 4.1 are used: ellipticity and boundedness of μ near y, which drive the local term; the uniform bound (A4), which drives domination; and enough global control to make the tail negligible. Global uniform ellipticity is therefore stronger than necessary. It can be replaced by ellipticity on $B(y, \delta_0)$ together with any condition forcing the contribution of $\{Z_h \notin B(y, \delta)\}$ to vanish after multiplication by $h^{(r-1)K/2} \sup_z g_h(y \mid z)^r$, for instance a Gaussian upper bound of Aronson type on r_h. We state the global version because it keeps the proof short and because the content of the theorem is the dimensional constant, not the weakest hypotheses.

We do not claim the assumption set is directly verifiable. Conditions (A1) to (A4) are checkable from the coefficients, but (A5) is not a primitive condition: it requires uniform convergence on a ball of the *Euler* density at time $1-h$ to the *diffusion* density at time 1, which bundles a discretisation limit together with continuity of $t \mapsto p_t$ in the uniform topology. Results of Bally and Talay (1996a) and Bally and Talay (1996b) deliver statements of this kind, but under hypotheses of their own that are stronger than (A1) and (A2), so appealing to them to justify (A5) while citing (A1) and (A2) as the primitive conditions would be mildly circular. The honest description is that (A5) is an assumption imported from the Euler-approximation literature, not one the reader can check from μ and V alone.

4.1 The scaling law without the convergence assumption

The objection to (A5) recorded in Remark 4.5 is real, and it bears on how much of this paper rests on an imported hypothesis. This subsection answers it: the *scale* $h^{-(r-1)K/2}$, which is what the paper's thesis depends on, follows from conditions checkable in μ and V, and (A5) is needed only to identify the constant.

The substitute is a two-sided Gaussian estimate of the kind Aronson proved for uniformly elliptic operators, transferred to the Euler density.

Assumption 4.6 (Aronson-type bounds on the preterminal density)**.** *There are constants* $0 < c_- \le c_+ < \infty$ *and* $0 < a_- \le a_+ < \infty$ *such that for every* $h \in (0, \frac{1}{2}]$ *and every* $z \in \mathbb{R}^K$,

$$c_-\,\varphi_K(z - x; 0, a_- I_K) \;\le\; r_h(z \mid x) \;\le\; c_+\,\varphi_K(z - x; 0, a_+ I_K). \tag{33}$$

Bounds of this form for Euler densities are established by Lemaire and Menozzi (2010) under conditions comparable to (A1) and (A2); the Gaussian upper bound alone goes back to Bally and Talay (1996b), and the parametrix expansions of Konakov and Mammen (2002) give related density estimates. We cite these as literature-based claims rather than reproving them, and the preterminal time $1 - h \ge \frac{1}{2}$ is bounded away from zero, which is what allows the constants to be taken uniform in h. The point of Assumption 4.6 is what it does *not* say: it implies (A3) and (A4), and it asserts nothing about convergence of r_h to anything.

Proposition 4.7 (Scale without (A5))**.** *Assume (A1), (A2) and Assumption 4.6, and let* $r \ge 1$. *Then*

$$\begin{aligned} c_-\,\varphi_K(y - x; 0, a_- I_K)\,\kappa_{r,K}(y) &\le \liminf_{h \downarrow 0} h^{(r-1)K/2}\mathbb{E}[G_h^r] \\ &\le \limsup_{h \downarrow 0} h^{(r-1)K/2}\mathbb{E}[G_h^r] \le c_+\,\varphi_K(y - x; 0, a_+ I_K)\,\kappa_{r,K}(y), \end{aligned} \tag{34}$$

where $\kappa_{r,K}(y) = c_{r,K}|V(y)|^{-(r-1)/2}$ *is the constant of Theorem 4.2 with* $p(y \mid x)$ *removed. In particular*

$$0 < \liminf_{h \downarrow 0} h^{(r-1)K/2}\mathbb{E}[G_h^r] \le \limsup_{h \downarrow 0} h^{(r-1)K/2}\mathbb{E}[G_h^r] < \infty, \tag{35}$$

so $\mathbb{E}[G_h^r] \asymp h^{-(r-1)K/2}$ *exactly, and the effective size of Definition 2.1 is unchanged.*

Proof. The proof of Theorem 4.2 splits $h^{(r-1)K/2}\mathbb{E}[G_h^r]$ into the contribution of $B(y, \delta)$ and its complement, and shows the second vanishes using only (A1), (A2) and a uniform bound on r_h, all of which are available here. For the local part, the same substitution $z = y - \sqrt{h}\,u + O(h)$ that produces (29) gives

$$h^{(r-1)K/2} \int_{\mathbb{R}^K} g_h(y \mid z)^r \, dz \longrightarrow \kappa_{r,K}(y),$$

since the powers of h cancel as in that proof, the Gaussian integral being $(2\pi/r)^{K/2}|V(y)|^{1/2}$. On the ball, bound r_h above by c_+ times the supremum of $\varphi_K(\cdot - x; 0, a_+ I_K)$ there, and below by c_- times the infimum of $\varphi_K(\cdot - x; 0, a_- I_K)$. Letting $\delta \downarrow 0$, both extrema converge to the value at y by continuity of the Gaussian density, which gives (34). For the lower bound no tail estimate is needed, every integrand being nonnegative, so it suffices to restrict the integral to $B(y, \delta)$. ☐

The bracket in (34) is wide, and necessarily so. Assumption 4.6 is required to hold uniformly over $h \in (0, \frac{1}{2}]$, so its constants must accommodate the whole range of preterminal variances at once. In the Brownian case, where r_h is exactly the $\mathcal{N}(x, (1 - h)I_K)$ density, that range is $[\frac{1}{2}, 1)$, which forces $a_- \le \frac{1}{2}$ and $a_+ \ge 1$; the admissible constants are then $c_- = a_-^{K/2}$ and $c_+ = (2a_+)^{K/2}$, and the resulting bracket contains $p(y \mid x)$ with room to spare. A bound calibrated at one value of h would be narrower but would not dominate the limit density, which is what the bracket has to contain. Nothing here claims tightness; the claim is containment together with strict positivity and finiteness of both ends.

Proposition 4.7 is the answer to the question of how much (A5) is doing. It is doing exactly one thing: replacing the bracket $[c_-\varphi_K(y-x;0,a_-I_K),\, c_+\varphi_K(y-x;0,a_+I_K)]$ by the single value $p(y \mid x)$. The exponent of h, and with it the effective size $R_{M,S} = S/M^{K/2}$, the variance scale of Corollary 4.10, and the incompatibility between the published rate condition and mean-square consistency in four dimensions, are all consequences of (35) alone and hold under conditions a reader can check from μ and V.

Remark 4.8 (What the proof actually consumes from (A5))**.** Even for the constant, (A5) is stronger than needed. The local part of the proof integrates r_h against a kernel of bandwidth $\sqrt{h/r}$ centred at y, so what is used is convergence along the rescaled neighbourhood only:

$$r_h(y-\sqrt{h}\,u \mid x) \longrightarrow p(y \mid x) \qquad \text{for almost every } u \in \mathbb{R}^K, \tag{36}$$

together with the domination already supplied by (A4). Uniform convergence on a fixed ball $\overline{B}(y,\delta_0)$ implies (36) and is strictly stronger, since (36) constrains r_h only on a ball of radius shrinking like $\sqrt{h}$. We keep (A5) in the statement of Assumption 4.1 because it is the form in which the Euler-approximation literature supplies such results, but (36) is the honest description of the hypothesis, and it is what a future primitive verification would have to establish.

Remark 4.9 (Time-inhomogeneous coefficients)**.** If μ and V depend on time, the one-step kernel (3) evaluates them at time $1-h$, and the proof goes through verbatim provided μ and V are jointly continuous at $(y,1)$ and (A1) and (A2) hold uniformly in t on a neighbourhood of 1. The limit constant is unchanged and $V(y)$ is read as $V(y,1)$. We suppress the time argument in the statement because carrying it obscures the cancellation of powers of h, which is where the dimensional constant comes from.

Corollary 4.10 (Generic variance scale)**.** *Let $q_M(y \mid x) = \mathbb{E}[G_h]$ and suppose $q_M(y \mid x) \to p(y \mid x)$. Then*

$$\operatorname{Var}(\widehat{q}_{M,S}) \sim \frac{\tau^2(x,y)}{Sh^{K/2}}, \qquad \tau^2(x,y) = (4\pi)^{-K/2} p(y \mid x)|V(y)|^{-1/2}. \tag{37}$$

The central limit theorem below centres at the Euler density $q_M(y \mid x) = \mathbb{E}[G_h]$, and its proof needs that quantity to be bounded uniformly in h. That is not automatic: the summand G_h itself is unbounded, since a Gaussian of covariance hV has height of order $h^{-K/2}$ at its centre. What makes the *expectation* bounded is that the preterminal state has a uniformly bounded density, so the mass the kernel can concentrate on is limited. We record this separately rather than assume it.

Lemma 4.11 (The Euler density is bounded uniformly in the step)**.** *Under Assumption 4.1,*

$$\sup_{0<h\le 1/2} q_M(y \mid x) \;\le\; \bar{r}\left(\frac{4\overline{v}}{\underline{v}}\right)^{K/2} + C_K\,\bar{r}\,\underline{v}^{-K/2}\bar{\mu}^K \;<\; \infty, \tag{38}$$

where C_K depends only on K. In particular $q_M(y \mid x) = O(1)$.

Proof. Writing Z_h for the preterminal state and substituting $u = y - z$, which has unit Jacobian,

$$q_M(y \mid x) = \int \varphi_{hV(y-u)}(u - \mu(y-u)h)\, r_h(y-u \mid x)\, du.$$

By (A4) the density factor is at most $\bar{r}$, and by (A2),

$$\varphi_{hV(z)}(w) \leq (2\pi h)^{-K/2}\underline{v}^{-K/2}\exp\Big(-\frac{\|w\|^2}{2h\overline{v}}\Big).$$

Split the integral at $\|u\| = 2\bar{\mu}h$. On $\|u\| \geq 2\bar{\mu}h$, (A1) gives $\|u-\mu(y-u)h\| \geq \|u\|-\bar{\mu}h \geq \|u\|/2$, so that part is at most

$$\bar{r}\,(2\pi h)^{-K/2}\underline{v}^{-K/2}\int_{\mathbb{R}^K}\exp\Big(-\frac{\|u\|^2}{8h\overline{v}}\Big)du = \bar{r}\,(2\pi h)^{-K/2}\underline{v}^{-K/2}\big(8\pi h\overline{v}\big)^{K/2} = \bar{r}\left(\frac{4\overline{v}}{\underline{v}}\right)^{K/2},$$

which is free of h. On $\|u\| < 2\bar{\mu}h$ the integrand is at most $\bar{r}(2\pi h)^{-K/2}\underline{v}^{-K/2}$ and the region has volume $\omega_K(2\bar{\mu}h)^K$, so that part is at most $C_K\bar{r}\underline{v}^{-K/2}\bar{\mu}^K h^{K/2}$ with $C_K = \omega_K 2^K(2\pi)^{-K/2}$, and $h \leq 1/2$ bounds it. Adding the two gives (38). □

Remark 4.12 (Boundedness, not convergence, is what is used)**.** Lemma 4.11 gives only $q_M = O(1)$, and that is all Theorem 4.13 requires: the theorem centres at q_M rather than at p, so no rate of approach is needed. The stronger statement $q_M(y \mid x) \to p(y \mid x)$ also holds under Assumption 4.1, by the approximate-identity argument of Theorem 4.2 applied at $r = 1$ together with (A5), but we do not use it here. Where the distinction matters is Corollary 4.15, which recentres at p and there assumes a first-order Euler density expansion separately.

Theorem 4.13 (Generic endpoint CLT)**.** *Let (M_n, S_n) satisfy $M_n \to \infty$ and $S_n \to \infty$, write $h_n = 1/M_n$, and suppose the effective endpoint size diverges:*

$$R_n = S_n h_n^{K/2} = \frac{S_n}{M_n^{K/2}} \longrightarrow \infty. \tag{39}$$

Then under Assumption 4.1,

$$\sqrt{R_n}\,\{\widehat{q}_{M_n,S_n}(y \mid x) - q_{M_n}(y \mid x)\} \Longrightarrow \mathcal{N}\Big(0, \tau^2(x,y)\Big), \tag{40}$$

where, as in (37)*,*

$$\tau^2(x,y) = (4\pi)^{-K/2}p(y \mid x)|V(y)|^{-1/2}. \tag{41}$$

This is a statement about the simulated transition density, centred at the Euler density q_{M_n} and not at the diffusion density p; Corollary 4.15 treats the latter. It is not a statement about any parameter estimator.

Proof. Define the triangular array

$$G_{n,s} = g_{h_n}(y \mid Z_{h_n,s}), \qquad s = 1,\ldots,S_n,$$

where $Z_{h_n,1},\ldots,Z_{h_n,S_n}$ are the independent preterminal draws in (5). Within row n these are independent and identically distributed with common mean $\mathbb{E}G_{n,1} = q_{M_n}(y \mid x)$, and

$$\widehat{q}_{M_n,S_n} - q_{M_n} = \frac{1}{S_n}\sum_{s=1}^{S_n}(G_{n,s} - \mathbb{E}G_{n,s}).$$

Variance. Theorem 4.2 with $r = 2$ gives $h_n^{K/2}\mathbb{E}[G_{n,1}^2] \to \tau^2$. Since $\mathbb{E}G_{n,1} = q_{M_n}$ is bounded uniformly in n by Lemma 4.11, $h_n^{K/2}(\mathbb{E}G_{n,1})^2 \to 0$, so

$$h_n^{K/2}\,\mathrm{Var}(G_{n,1}) \longrightarrow \tau^2, \qquad \mathrm{Var}(\widehat{q}_{M_n,S_n}) = \frac{\mathrm{Var}(G_{n,1})}{S_n} \sim \frac{\tau^2}{R_n}. \tag{42}$$

Third centred moment. Theorem 4.2 with $r = 3$ bounds the *raw* moment, $\mathbb{E}[G_{n,1}^3] = O(h_n^{-K})$. The centred moment does not follow from this automatically, and is obtained as follows. The summand is a Gaussian density value, hence nonnegative, so the elementary inequality $|a - b|^3 \le 4(a^3 + b^3)$ for $a, b \ge 0$ applies and

$$\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3 \le 4\left\{\mathbb{E}[G_{n,1}^3] + (\mathbb{E}G_{n,1})^3\right\} = O(h_n^{-K}) + O(1) = O(h_n^{-K}), \tag{43}$$

using Lemma 4.11 once more. Nonnegativity of the summand is what makes this step elementary; without it the centred third moment would require a separate argument.

Lyapunov condition. Write $\sigma_n^2 = \mathrm{Var}(G_{n,1})$, so $\sigma_n^2 \sim \tau^2 h_n^{-K/2}$ by (42). The Lyapunov ratio at order $2 + \delta$ with $\delta = 1$ is

$$L_n = \frac{S_n\,\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3}{(S_n\sigma_n^2)^{3/2}} = \frac{S_n\,O(h_n^{-K})}{S_n^{3/2}\,\tau^3 h_n^{-3K/4}\{1 + o(1)\}} = O\left(S_n^{-1/2}h_n^{-K/4}\right) = O\left(R_n^{-1/2}\right),$$

because $S_n^{-1/2}h_n^{-K/4} = (S_n h_n^{K/2})^{-1/2}$. Condition (39) gives $L_n \to 0$.

The Lyapunov central limit theorem for triangular arrays with row-wise independent summands therefore applies to $(S_n\sigma_n^2)^{-1/2}\sum_s(G_{n,s} - \mathbb{E}G_{n,s})$. Multiplying by $\sigma_n h_n^{K/4} \to \tau$ converts the normalisation $\sqrt{S_n}\,\sigma_n$ into $\sqrt{R_n}$ and yields (40). □

Remark 4.14 (Why a fixed-M central limit theorem cannot be reused)**.** For fixed M the summands do not change with S, their variance is a constant, and the classical $\sqrt{S}$ central limit theorem applies. The array above is genuinely triangular: the law of $G_{n,s}$ changes with n through h_n, and its variance diverges at rate $h_n^{-K/2}$. The correct normalisation is $\sqrt{R_n}$, which coincides with $\sqrt{S_n}$ only when $K = 0$. Substituting a fixed-M central limit theorem and then letting M grow, as in Lemma 3 of Brandt and Santa-Clara (2002), is exactly the step that fails; the limiting variance stated there is the variance of a summand that itself depends on M.

Corollary 4.15 (CLT centred at the diffusion density)**.** *Suppose in addition that*

$$q_M(y \mid x) - p(y \mid x) = b(x, y)h + o(h). \tag{44}$$

If

$$Sh^{K/2} \to \infty, \qquad \sqrt{Sh^{K/2}}\,h \to 0, \tag{45}$$

then

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\} \Longrightarrow \mathcal{N}(0, \tau^2(x, y)). \tag{46}$$

In terms of M and S, the second condition in (45) *is*

$$\frac{\sqrt{S}}{M^{1+K/4}} \longrightarrow 0. \tag{47}$$

For $K = 4$, the two conditions together define the feasible regime

$$M^2 \ll S \ll M^4. \tag{48}$$

Proof. Decompose

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S} - p\} = \sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S} - q_M\} + \sqrt{Sh^{K/2}}\,\{q_M - p\}\,.$$

The first term converges to $\mathcal{N}(0, \tau^2)$ by Theorem 4.13. By (44) the second term equals

$$\sqrt{Sh^{K/2}}\,\{b(x, y)h + o(h)\}\,,$$

which vanishes precisely under the second condition in (45). Slutsky's theorem gives (46).

For the restatement, $\sqrt{Sh^{K/2}}\,h = \sqrt{S}\,h^{K/4+1} = \sqrt{S}/M^{1+K/4}$, since $h = 1/M$. This is (47), and squaring gives the equivalent form $S/M^{2+K/2} \to 0$.

For $K = 4$ the first condition in (45) reads $S/M^2 \to \infty$ and the second reads $S/M^4 \to 0$, which is (48). The regime is nonempty: $S = M^3$ satisfies both. $\square$

Remark 4.16 (Bias negligibility is not variance consistency)**.** The two conditions in (45) do different work and pull in opposite directions. The first is $S/M^{K/2} \to \infty$: it makes the Monte Carlo error vanish and is what Theorem 4.13 needs. It is a *lower* bound on S, requiring $S \gg M^{K/2}$. The second is $S/M^{2+K/2} \to 0$, equivalently $\sqrt{S}/M^{1+K/4} \to 0$: it makes the Euler bias negligible *relative to the Monte Carlo standard deviation*, and because that standard deviation shrinks as S grows, it is an *upper* bound on S, requiring $S \ll M^{2+K/2}$. At $K = 4$ the pair is $M^2 \ll S \ll M^4$. Conflating the two is the source of the difficulty in Table 3: the published condition $\sqrt{S}/M \to 0$, that is $S \ll M^2$, has the form of the second requirement but at the wrong power, and it then contradicts the first when $K \geq 4$.

Remark 4.17 (Dependency structure)**.** The density-level results depend on one another as follows. Assumption 4.1 supports Theorem 4.2, which is used only through its cases $r = 2$ and $r = 3$. The case $r = 2$ gives the variance scale Corollary 4.10; the two cases together give the Lyapunov condition in Theorem 4.13. Corollary 4.15 adds the Euler bias expansion (44), which is an assumption imported from Bally and Talay (1996a) and Bally and Talay (1996b) rather than proved here, and combines it with Theorem 4.13. Proposition 5.1 uses Corollary 4.10 and (44) but no central limit theorem. In the Brownian benchmark, Proposition 3.1 replaces Theorem 4.2 with an exact computation and the bias expansion is unnecessary because the bias is zero, so Theorem 3.16 rests on no imported result at all. Corollary 3.7 is independent of every central limit theorem above; it uses only the exact Gaussian tail bound.

Remark 4.18 (The dimensional conflict)**.** The condition $\sqrt{S}/M \to 0$ suppresses an $O(M^{-1})$ Euler bias under an incorrectly fixed Monte Carlo scale. Once the variance inflation $M^{K/2}$ is included, the bias must be compared with the standard deviation $M^{K/4}/\sqrt{S}$, producing (47). In four dimensions, the lower consistency requirement and the upper bias requirement leave the nonempty interval (48). The published condition places S below the lower boundary of that interval.

5 Bias–variance trade-off and dimensional rate

Suppose the Euler density admits (44). Because (5) is unbiased for q_M,

$$\begin{aligned}\mathbb{E}\Big[(\widehat{q}_{M,S} - p)^2\Big] &= (q_M - p)^2 + \mathrm{Var}(\widehat{q}_{M,S}) \\ &= b(x,y)^2 h^2 + \frac{\tau^2(x,y)}{Sh^{K/2}} + o\Big(h^2 + \frac{1}{Sh^{K/2}}\Big). \end{aligned} \tag{49}$$

Proposition 5.1 (Pointwise optimal discretisation)**.** *If $b(x,y) \neq 0$, the leading terms in* (49) *are balanced by*

$$h \asymp S^{-2/(K+4)}, \qquad M \asymp S^{2/(K+4)}. \tag{50}$$

The resulting optimal pointwise mean squared error is

$$\mathrm{MSE}_{\mathrm{opt}} \asymp S^{-4/(K+4)}. \tag{51}$$

Proof. Differentiate the leading expression $Bh^2 + C/(Sh^{K/2})$ with respect to h or equate its two terms. This gives $h^{K/2+2} \asymp S^{-1}$ and hence (50). Substitution yields (51). $\square$

The rate is the familiar structure of a kernel-density problem: reducing the bandwidth controls approximation bias but decreases the effective number of simulations. The dimension enters through the volume of the final Gaussian kernel.

Remark 5.2 (The optimal design is not a design at which the CLT centres at p)**.** The mean-squared-error optimum of Proposition 5.1 and the recentred central limit theorem of Corollary 4.15 pull in opposite directions, and the tension is worth naming rather than leaving for a reader to trip over. At $M \asymp S^{2/(K+4)}$ the two error sources are balanced by construction, so the bias-negligibility condition required for a limit centred at p, namely $\sqrt{S}/M^{1+K/4} \to 0$, fails exactly there. The consequence is the usual one in nonparametric estimation: at the mean-squared-error optimum the limiting distribution acquires a mean shift, and a confidence interval built from the centred limit needs undersmoothing, that is a smaller h than the optimal one, to remain valid. Neither statement is wrong; they answer different questions, one about point accuracy and one about coverage.

Some illustrative exponents are

Dimension K	M_{opt}	Optimal MSE
1	$S^{2/5}$	$S^{-4/5}$
2	$S^{1/3}$	$S^{-2/3}$
4	$S^{1/4}$	$S^{-1/2}$
8	$S^{1/6}$	$S^{-1/3}$

This does not mean that M should always be chosen by pointwise MSE. Likelihood estimation requires accuracy over many endpoints and parameter values, and higher-order or bridge-based methods can change both constants and rates. It does mean that increasing M while holding S fixed, or while letting S grow too slowly relative to dimension, can make the simulation approximation worse rather than better.

Table 3: Principal endpoint-density rate conditions.

Property	General dimension K	Four dimensions
Monte Carlo mean-square consistency	$S/M^{K/2} \to \infty$	$S/M^2 \to \infty$
Density CLT centred at q_M	$S/M^{K/2} \to \infty$	$S/M^2 \to \infty$
Euler bias negligible in corrected CLT	$\sqrt{S}/M^{1+K/4} \to 0$	$S/M^4 \to 0$
Condition in Brandt and Santa-Clara (2002)	$\sqrt{S}/M \to 0$	$S/M^2 \to 0$

6 From density error to log-likelihood error

For observations $Y_0, \ldots, Y_N$, the simulated log likelihood is a sum of terms

$$\log \widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta).$$

Even when $\widehat{q}_{M,S}$ is unbiased for q_M, its logarithm is not unbiased. The effective endpoint size $R_{M,S} = Sh^{K/2}$ controls this nonlinearity.

The next result is conditional. Its first two statements follow from Theorem 4.13, but the third holds only under an additional uniform-integrability and lower-tail condition, stated in the proposition and assumed rather than verified.

Proposition 6.1 (Log-density delta expansion)**.** *Fix an endpoint with $q_M \to p > 0$ and suppose the conditions of Theorem 4.13 hold. If $R_{M,S} = Sh^{K/2} \to \infty$, then*

$$\sqrt{R_{M,S}}\,\{\log \widehat{q}_{M,S} - \log q_M\} \Longrightarrow \mathcal{N}\left(0, \frac{\tau^2}{p^2}\right), \tag{52}$$

and

$$\log \widehat{q}_{M,S} - \log q_M = \frac{\widehat{q}_{M,S} - q_M}{q_M} - \frac{(\widehat{q}_{M,S} - q_M)^2}{2q_M^2} + o_{\mathbb{P}}(R_{M,S}^{-1}). \tag{53}$$

Whenever the second-order expansion is uniformly integrable, it follows that

$$\mathbb{E}[\log \widehat{q}_{M,S}] - \log q_M = -\frac{\tau^2}{2p^2 R_{M,S}} + o(R_{M,S}^{-1}). \tag{54}$$

Proof. The first statement is the delta method applied to Theorem 4.13. Since $(\widehat{q} - q_M)/q_M = O_{\mathbb{P}}(R^{-1/2})$ and q_M is bounded away from zero, Taylor expansion of $\log(1+u)$ gives (53). Taking expectations requires uniform integrability of the remainder and lower-tail control; under that additional condition, the first-order term has zero expectation and the second-order term gives (54). □

The leading nonlinear scale is

$$R_{M,S}^{-1} = \frac{M^{K/2}}{S}, \tag{55}$$

not uniformly $1/S$. The statement in Brandt and Santa-Clara (2002, p. 171) that the logarithmic transformation induces a bias of order $1/S$ is valid with M fixed. It is not a joint-M, S order.

Suppose simulations are independent across observations and the local constants are uniformly bounded. For the average simulated criterion

$$\widehat{Q}_{N,M,S}(\theta) = \frac{1}{N}\sum_{n=0}^{N-1} \log \widehat{q}_{n,M,S}(\theta),$$

the stochastic average of the first-order simulation error has standard deviation of order

$$(NR_{M,S})^{-1/2},$$

while the average nonlinear bias remains of order $R_{M,S}^{-1}$. The simulation bias does not average out across observations.

A full estimator theorem requires uniform control in θ of densities, scores, and Hessians. The following statement is an abstract *local* perturbation result. It is not a theorem about the simulated-likelihood estimator: it assumes both maximisers are localised near θ_0 and assumes the uniform score and Hessian bounds, none of which we verify for that estimator. It is included to separate what the endpoint calculations deliver from what an estimator theorem would additionally require. Every hypothesis below is probabilistic, so the conclusion is a statement in probability, not a deterministic implication.

Proposition 6.2 (Abstract local equivalence of exact and simulated maximisers)**.** *Let $U \subseteq \boldsymbol{\Theta}$ be a fixed open neighbourhood of θ_0, let Q_N be an exact average log likelihood and $\widehat{Q}_{N,M,S}$ a simulated criterion, both twice continuously differentiable on U. Let $\widehat{\theta}_N$ be an interior local maximiser of Q_N in U and $\widetilde{\theta}_{N,M,S}$ an interior local maximiser of $\widehat{Q}_{N,M,S}$ in U. Assume:*

(i) both maximisers are consistent: $\widehat{\theta}_N \to \theta_0$ and $\widetilde{\theta}_{N,M,S} \to \theta_0$, in probability;

(ii) with probability tending to one both first-order conditions hold, $\nabla Q_N(\widehat{\theta}_N) = 0$ and $\nabla \widehat{Q}_{N,M,S}(\widetilde{\theta}_{N,M,S}) = 0$;

(iii) the scores agree uniformly on U at the parametric rate,

$$\sup_{\theta \in U} \|\nabla \widehat{Q}_{N,M,S}(\theta) - \nabla Q_N(\theta)\| = o_{\mathbb{P}}(N^{-1/2});$$

(iv) the Hessians agree uniformly on U,

$$\sup_{\theta \in U} \|\nabla^2 \widehat{Q}_{N,M,S}(\theta) - \nabla^2 Q_N(\theta)\| = o_{\mathbb{P}}(1);$$

(v) the exact Hessian converges uniformly on U in probability to a continuous matrix function H, with $H(\theta_0)$ nonsingular;

(vi) with probability tending to one the line segment joining $\widehat{\theta}_N$ and $\widetilde{\theta}_{N,M,S}$ lies in U.

Then

$$\sqrt{N}(\widetilde{\theta}_{N,M,S} - \widehat{\theta}_N) \to 0 \quad \textit{in probability.}$$

Proof. Work on the event, of probability tending to one, on which (ii) and (vi) both hold. Write $\Delta_N = \widetilde{\theta}_{N,M,S} - \widehat{\theta}_N$. Since $\widehat{Q}_{N,M,S}$ is twice continuously differentiable on U and the segment lies in U, the fundamental theorem of calculus applied to $s \mapsto \nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N + s\Delta_N)$ gives

$$0 = \nabla \widehat{Q}_{N,M,S}(\widetilde{\theta}_{N,M,S}) = \nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N) + \bar{H}_N \Delta_N, \qquad \bar{H}_N = \int_0^1 \nabla^2 \widehat{Q}_{N,M,S}(\widehat{\theta}_N + s\Delta_N)\, ds.$$

The integral form is used rather than a coordinatewise mean-value theorem, which would supply a different intermediate point in each row and no single matrix.

For the right-hand side, the exact first-order condition $\nabla Q_N(\widehat{\theta}_N) = 0$ gives

$$\|\nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N)\| = \|\nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N) - \nabla Q_N(\widehat{\theta}_N)\| \le \sup_{\theta \in U} \|\nabla \widehat{Q}_{N,M,S}(\theta) - \nabla Q_N(\theta)\| = o_{\mathbb{P}}(N^{-1/2})$$

by (iii).

For the matrix, decompose

$$\begin{aligned} \|\bar{H}_N - H(\theta_0)\| \le & \sup_{\theta \in U} \|\nabla^2 \widehat{Q}_{N,M,S}(\theta) - \nabla^2 Q_N(\theta)\| + \sup_{\theta \in U} \|\nabla^2 Q_N(\theta) - H(\theta)\| \\ & + \sup_{0 \le s \le 1} \|H(\widehat{\theta}_N + s\Delta_N) - H(\theta_0)\|. \end{aligned}$$

The first term is $o_{\mathbb{P}}(1)$ by (iv) and the second by (v). For the third, (i) makes both endpoints of the segment converge to θ_0 in probability, so $\sup_{0 \le s \le 1} \|\widehat{\theta}_N + s\Delta_N - \theta_0\| \to 0$ in probability; continuity of H at θ_0 then makes the term $o_{\mathbb{P}}(1)$. Hence $\bar{H}_N \to H(\theta_0)$ in probability. Because $H(\theta_0)$ is nonsingular and matrix inversion is continuous at a nonsingular point, with probability tending to one $\bar{H}_N$ is invertible and $\bar{H}_N^{-1} = O_{\mathbb{P}}(1)$.

Combining, $\Delta_N = -\bar{H}_N^{-1} \nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N)$, so

$$\sqrt{N}\, \Delta_N = -\bar{H}_N^{-1} \cdot \sqrt{N}\, \nabla \widehat{Q}_{N,M,S}(\widehat{\theta}_N) = O_{\mathbb{P}}(1) \cdot o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1). \qquad \square$$

Remark 6.3 (Where the localisation would have to come from)**.** Assumption (i) assumes consistency of the simulated maximiser rather than deriving it, and it is the hypothesis that does the most work: without it the segment need not shrink to θ_0, $\bar{H}_N$ need not approach a nonsingular limit, and the argument fails. It could be replaced by the weaker requirement $\mathbb{P}(\widetilde{\theta}_{N,M,S} \in U) \to 1$ if H were additionally assumed nonsingular throughout U, with smallest singular value bounded away from zero there. Either way the localisation must be supplied, and the usual routes are global uniform convergence of the simulated criterion to a limit criterion uniquely maximised at θ_0, or an argmax theorem. We prove neither here, and Section 7.7 explains why the transition-density counterexample does not by itself settle that question. Proposition 6.2 should therefore be read as a conditional abstract result, not as an asymptotic theorem for the endpoint simulated-likelihood estimator.

One of those requirements can be made exact rather than plausible, and it is the one that matters most. Hypothesis (iii) of Proposition 6.2 is about the *score*, not the density, and by Proposition A.1 the simulated density is a kernel estimate of bandwidth $\sqrt{h}$; differentiating a kernel estimate costs one further power of the bandwidth. For the Brownian benchmark the cost can be computed in closed form, and the answer is clean.

Proposition 6.4 (Exact score second moment)**.** *For K-dimensional Brownian motion at $x = y = 0$, let G_h be the endpoint summand* (8) *and let ∂_j denote differentiation in the j-th coordinate of the evaluation point. Then for every $h \in (0, 1)$ and every j,*

$$\mathbb{E}\big[(\partial_j G_h)^2\big] = (1-h)(2\pi)^{-K} h^{-K/2-1}(2-h)^{-K/2-1} = \frac{1-h}{h(2-h)}\,\mathbb{E}\big[G_h^2\big]. \tag{56}$$

The ratio to the density second moment is therefore free of K, and as $h \downarrow 0$,

$$\mathbb{E}\big[(\partial_j G_h)^2\big] \;\sim\; (2\pi)^{-K} 2^{-(K+2)/2}\, h^{-(K+2)/2}. \tag{57}$$

Proof. At $y = x = 0$ the preterminal draw is $Z = \sqrt{1-h}\,\xi$ with $\xi \sim \mathcal{N}(0, I_K)$, and $\partial_j G_h = (Z_j/h)G_h$. Hence

$$\mathbb{E}\big[(\partial_j G_h)^2\big] = h^{-2}\,\mathbb{E}\big[Z_j^2 G_h^2\big] = h^{-2}(1-h)(2\pi h)^{-K}\,\mathbb{E}\big[\xi_j^2 e^{-a\|\xi\|^2}\big], \qquad a = \frac{1-h}{h}.$$

Writing $c = a + \frac{1}{2} = (2-h)/(2h)$ and integrating, $\mathbb{E}[\xi_j^2 e^{-a\|\xi\|^2}] = (2c)^{-K/2-1} = (h/(2-h))^{K/2+1}$. Substituting gives (56); the second equality follows from $\mathbb{E}[G_h^2] = (2\pi)^{-K} h^{-K/2}(2-h)^{-K/2}$, which is Proposition 3.1 at $r = 2$ and $y = x$. Letting $h \downarrow 0$ gives (57). ☐

The consequence is a *score-level* effective simulation size, one power of M more demanding than the density-level one:

$$R_{M,S}^{\text{score}} = S\,h^{(K+2)/2} = \frac{S}{M^{(K+2)/2}}, \qquad \text{against} \qquad R_{M,S} = \frac{S}{M^{K/2}}. \tag{58}$$

Before drawing the consequence, one identification must be made explicit rather than assumed, because Proposition 6.4 differentiates in the *evaluation point y* while hypothesis (iii) concerns ∇_θ.

Assumption 6.5 (Score identification)**.** *The model is such that differentiation of the summand in the parameter reduces to differentiation in the evaluation point, up to a bounded nonsingular linear map: $\nabla_\theta G_h = J\,\nabla_y G_h + \mathcal{R}_h$ with $\|J\|$ and $\|J^{-1}\|$ bounded and $\mathbb{E}\|\mathcal{R}_h\|^2 = o(h^{-(K+2)/2})$.*

The assumption is not vacuous, and it is not confined to the location model either. It holds exactly, with $\mathcal{R}_h \equiv 0$, for every constant-coefficient family whose diffusion matrix does not depend on the parameter.

Proposition 6.6 (The identification holds for constant-coefficient families)**.** *Let $dY_t = \mu(\theta)\,dt + \Sigma\,dW_t$ with $\mu : \boldsymbol{\Theta} \to \mathbb{R}^K$ continuously differentiable, Σ constant and nonsingular, and $V = \Sigma\Sigma^\mathsf{T}$ free of θ. Then for every h,*

$$\nabla_\theta G_h = -J(\theta)^\mathsf{T}\,\nabla_y G_h, \qquad J(\theta) = \frac{\partial \mu}{\partial \theta},$$

so Assumption 6.5 holds with $\mathcal{R}_h \equiv 0$ whenever J and J^{-1} are bounded on U.

Proof. Write the preterminal draw as $Z_h = x + \mu(\theta)(1-h) + \sqrt{1-h}\,\Sigma\xi$ with $\xi \sim \mathcal{N}(0, I_K)$, which is exact because the coefficients are constant. The summand evaluates the one-step density at $y - Z_h - \mu(\theta)h$, and

$$y - Z_h - \mu(\theta)h = y - x - \mu(\theta)(1-h) - \sqrt{1-h}\,\Sigma\xi - \mu(\theta)h = (y - x - \mu(\theta)) - \sqrt{1-h}\,\Sigma\xi.$$

The two occurrences of $\mu(\theta)$, one in the law of Z_h and one in the kernel mean, combine into a single shift that is free of h. Since V does not depend on θ, the parameter therefore enters G_h only through the argument $y - x - \mu(\theta)$, in which it appears exactly as y does but with the opposite sign. Differentiating and applying the chain rule gives the display, with $\partial/\partial\theta$ of the argument equal to $-J$ and $\partial/\partial y$ equal to the identity. $\square$

Remark 6.7 (Beyond the constant-coefficient case)**.** Three regimes should be kept apart. In the location model $dY = \theta\,dt + dW$ of Section A the identification is Proposition A.1 itself, with $J = -I$. In the constant-coefficient family of Proposition 6.6 it again holds exactly, which covers any parameterisation of a constant drift together with a fixed diffusion matrix, so Corollary 6.8 is a proved statement and not a conditional one for that class.

For a state-dependent drift the identification is genuinely weaker, and we do not claim it. Taking the Ornstein–Uhlenbeck process with $\theta = \alpha$ as the simplest example, the parameter enters the factor $(1 - \alpha h)$, the preterminal mean and the preterminal variance, so $\nabla_\alpha G_h = c_h(\xi)\nabla_y G_h$ with a *random* scalar $c_h(\xi)$ rather than a constant matrix. That scalar is $O(1)$ in h and has Gaussian-polynomial tails, so the exponent $h^{-(K+2)/2}$ of Proposition 6.4 should survive with a modified constant; but it changes the constant, it is not of the form assumed, and we have not carried the computation out. Outside Proposition 6.6 the score-level exponent should be read as expected rather than proved.

Corollary 6.8 (The published condition fails more decisively at score level)**.** *Let $K = 4$ and suppose Assumption 6.5 holds. Then hypothesis (iii) of Proposition 6.2 requires $S/M^3 \to \infty$, whereas the rate condition of Brandt and Santa-Clara (2002), equivalent to $S/M^2 \to 0$, requires $S \ll M^2$. Since $M^3 \gg M^2$, no sequence satisfies both, and the gap is wider by a factor M than the density-level incompatibility of Corollary 3.14.*

Proof. Hypothesis (iii) asks for $\sup_{\theta\in U}\|\nabla\widehat{Q}_{N,M,S} - \nabla Q_N\| = o_{\mathbb{P}}(N^{-1/2})$. Fix θ first. The simulated criterion is the average over n of $\log\widehat{q}_n$, and the N observations use independent simulation draws, so the pointwise discrepancy is an average of N independent terms each of standard deviation of order $(Sh^{(K+2)/2})^{-1/2}$ by Proposition 6.4 and Assumption 6.5, together with a bias term common across n. The stochastic part is therefore of order $N^{-1/2}(Sh^{(K+2)/2})^{-1/2}$, and requiring it to be $o_{\mathbb{P}}(N^{-1/2})$ gives $Sh^{(K+2)/2} \to \infty$, which at $K = 4$ is $S/M^3 \to \infty$. It is $Sh^{(K+2)/2} \to \infty$ rather than $\sqrt{N}/(Sh^{(K+2)/2}) \to 0$ that is needed here because averaging over n already supplies the factor $N^{-1/2}$; the bias term, which does not average away, is what produces the second condition in (59). Passing from the pointwise statement to the supremum over U requires a uniform law; we assume rather than prove it, as recorded in Remark 6.3, so the necessity direction is what the corollary uses and it holds pointwise already. The comparison with $S/M^2 \to 0$ is immediate. $\square$

This bears directly on Lemma 4 of Brandt and Santa-Clara (2002), which asserts $\ln\widehat{\mathcal{L}} - \ln\mathcal{L} = o(N/S^{1/2})$ under $S^{1/2}/M \to 0$ and derives it from Lemma 3, a statement about the density. The score is not the density, and Proposition 6.4 quantifies the difference exactly: at $K = 4$ the score's Monte Carlo variance is larger by a factor of order M. Whether the argmax conclusion survives is still open, for the reasons set out in Section A; what is not open is that the rate condition under which the published proof operates is insufficient for the score even to converge.

Under uniform analogues of the pointwise density expansion, the remaining requirements for first-order equivalence include

$$Sh^{(K+2)/2} \to \infty, \qquad \frac{\sqrt{N}}{Sh^{K/2}} \to 0, \qquad \sqrt{N}\, h \to 0, \tag{59}$$

corresponding respectively to vanishing score simulation noise, vanishing nonlinear simulation bias at the $N^{-1/2}$ scale, and negligible Euler bias. The first condition is the score-level one of (58), not the density-level $Sh^{K/2} \to \infty$; Proposition 6.4 is what fixes the exponent. Therefore (59) is a design implication, not a claimed complete theorem for arbitrary diffusions.

The broader Monte Carlo likelihood literature treats joint data and simulation limits through empirical-process, importance-sampling, or exact-simulation arguments rather than by combining two pointwise limits; see, among others, Lee (1995), Sung and Geyer (2007), Beskos, Papaspiliopoulos, and Roberts (2009), and Miasojedow et al. (2016).

7 What fails in the published asymptotic argument

This section distinguishes statements refuted by the Brownian example from steps that are merely unproved under the stated assumptions. Two published statements are directly refuted: the joint transition-density consistency of Lemma 2 and the joint central-limit statement of Lemma 3, by Corollary 3.7 and Corollary 3.11 respectively. Everything else below is a gap in a proof, not a demonstrated falsehood. Remark 3.12 states the hierarchy compactly.

That the distinction is worth drawing carefully is confirmed by how the results were received. Stramer and Yan (2007b), proposing the modified Brownian bridge as an alternative sampler, write that consistency and asymptotic normality for it "is conjectured to be true", but that "a rigorous proof, which would extend the results of Pedersen (1995) and Brandt and Santa-Clara (2002b), is not straightforward and is worth further investigation". The published theorems were thus taken as an established base to be extended rather than as statements to be checked, which is the ordinary and reasonable default. The purpose of this section is to identify precisely which parts of that base do not carry the weight.

7.1 The published statements

Because everything below turns on the quantifier structure of the published lemmas, we reproduce the two that are refuted, together with the sentence that fixes their role. All are from Appendix A of Brandt and Santa-Clara (2002, pp. 202–203), in their notation, with $\widehat{q}_{M,S}$ their simulated density, q_M the Euler density, and p the diffusion density.

The appendix opens: "Lemmas 1–6 combine into a proof of Theorem 1 and Lemma 7 establishes Theorem 2."

Lemma 2. *Given Assumptions 1 and 2, as $M \to \infty$ and $S \to \infty$,*

$$\widehat{q}_{M,S}(Y_{t_{n+1}}, t_{n+1} \mid Y_{t_n}, t_n; \theta) \to p(Y_{t_{n+1}}, t_{n+1} \mid Y_{t_n}, t_n; \theta) \quad \textit{almost surely.}$$

Lemma 3. *Given Assumptions 1 and 2, as $M \to \infty$ and $S \to \infty$, with $S^{1/2}/M \to 0$,*

$$\begin{aligned} S^{1/2}\,&[\widehat{q}_{M,S}(Y_{t_{n+1}}, t_{n+1} \mid Y_{t_n}, t_n; \theta) - p(Y_{t_{n+1}}, t_{n+1} \mid Y_{t_n}, t_n; \theta)] \\ &\sim N\big(0, \operatorname{var}[\phi(Y_{t_{n+1}}; z_s + \mu(z_s)h, V(z_s)h)]\big). \end{aligned}$$

Three features of these statements are what the results below engage with, and each can be read off the display.

First, **Lemma 2 carries no rate condition**. Its hypotheses are Assumptions 1 and 2 and the requirement that both indices diverge. Nothing restricts how they diverge relative to one another, so any sequence with $M_n \to \infty$ and $S_n \to \infty$ falls under it, $M = S$ among them. Corollary 3.7 exhibits such a sequence along which the stated conclusion fails.

Second, **Lemma 3 does carry the rate condition** $S^{1/2}/M \to 0$, and the sequence $M = S = n$ satisfies it, since $S^{1/2}/M = n^{-1/2} \to 0$. Corollary 3.11 therefore refutes Lemma 3 under its own stated hypothesis rather than outside it.

Third, the **dependency is explicit in the published proofs**, so the consequences propagate. Lemma 2's proof establishes $\widehat{q}_{M,S} \to q_M$ as $S \to \infty$ and then writes "Use of Lemma 1 completes the proof". Lemma 4's proof ends "The last equality follows from Lemma 3". Lemma 5's proof expands the simulated log likelihood and invokes Lemma 4 for the two differences. With the appendix's own summary sentence, the chain is

$$\text{Lemma 1+Lemma 2} \longrightarrow \text{Lemma 3} \longrightarrow \text{Lemma 4} \longrightarrow \text{Lemma 5} \xrightarrow{\text{+ Lemma 6}} \text{Theorem 1},$$

with Lemma 7 establishing Theorem 2. Refuting Lemmas 2 and 3 therefore removes inputs that the published proofs of Theorems 1 and 2 use, which is the precise sense in which those proofs fail. It is not the sense in which their conclusions are false; see Section 7.7.

7.2 Lemma 2: sequential convergence is used as joint convergence

The proof of Lemma 2 in Brandt and Santa-Clara (2002) applies the strong law to the Monte Carlo average and then invokes convergence of the Euler density. For every fixed M,

$$\widehat{q}_{M,S} \to q_M \quad \text{as } S \to \infty,$$

and separately

$$q_M \to p \quad \text{as } M \to \infty.$$

This gives the iterated limit

$$\lim_{M\to\infty} \lim_{S\to\infty} \widehat{q}_{M,S} = p.$$

It does not give convergence along every sequence $(M_n, S_n) \to (\infty, \infty)$. The summand changes with M and its variance diverges as $M^{K/2}$. Corollary 3.7 directly refutes the stated joint conclusion.

Two features of Lemma 2 are worth isolating. It is stated as a joint limit in M and S with no rate restriction whatsoever, so the refutation does not depend on the separate discussion of $\sqrt{S}/M \to 0$ below; and it is stated under Assumptions 1 and 2 alone, which standard Brownian motion satisfies. The counterexample therefore meets the lemma on its own terms.

A defender might reply that Lemma 2 was *intended* as the iterated statement, and that the joint phrasing is a slip of exposition rather than a claim. The reply does not save the chain, and the reason is visible in the published proofs rather than in any reconstruction of intent. Lemmas 4 and 5 are both stated "as $N \to \infty$, $M \to \infty$, and $S \to \infty$, with $S^{1/2}/M \to 0$", that is under a genuine joint limit with a rate tying the two indices together, and Lemma 4's proof invokes Lemma 3, whose proof in turn invokes Lemma 1 together with the same rate condition. An iterated reading of Lemma 2 would therefore be too weak to feed the arguments that use it: one cannot first send $S \to \infty$ at fixed M and then send $M \to \infty$ inside a statement in which S and M are constrained to move together. Either the lemma means what it says, in which case Corollary 3.7 refutes it, or it means the iterated statement, in which case Lemmas

4 and 5 rest on an input weaker than they require. The chain fails on either reading, which is why the quantifier question, though worth settling, does not decide the outcome.

Because Lemmas 3 to 6 take Lemma 2 as an input, and Theorem 1 is stated as the summary of Lemmas 1 to 6, the published proofs of Theorems 1 and 2 fail. That is a statement about the proofs. It does not establish that the conclusions of those theorems are false, and the subsections that follow identify further steps that are independently unproved.

7.3 Lemma 3: the joint central-limit statement is directly false

The paper uses a $\sqrt{S}$ central limit theorem and requires $\sqrt{S}/M \to 0$ to suppress the Euler bias. Two objections apply, and they are of different strengths.

The weaker objection is that the statement is ill-posed. The variance of the triangular-array summand is not asymptotically constant; it is of order $M^{K/2}$, so the limiting variance written in Lemma 3, namely the variance of $\phi(Y_{t_{n+1}}; z_s + \mu h, Vh)$, is the variance of a quantity that itself depends on M and diverges. The correct standardisation is $\sqrt{S/M^{K/2}}$, and the Euler bias must then be compared with that scale, producing (47).

The stronger objection is that the statement is false, not merely ill-posed. Corollary 3.11 shows that along $M_n = S_n = n$, a sequence satisfying the stated rate condition, $\sqrt{S_n}(\widehat{q}_{n,n} - p) \to -\infty$ in probability. The sequence is not tight, so it converges in distribution to no limit at all, and no choice of limiting variance rescues the statement. This refutation is independent of the objection about ill-posedness and does not rely on identifying what the intended limiting variance was.

7.4 Lemma 4: an $O_{\mathbb{P}}$ term is treated as $o_{\mathbb{P}}$

Let $x_n = \widehat{q}_n - p_n$. A nondegenerate central limit theorem implies $x_n = O_{\mathbb{P}}(S^{-1/2})$ for fixed M, not $o_{\mathbb{P}}(S^{-1/2})$. The Taylor expansion

$$\log\left(1 + \frac{x_n}{p_n}\right) = \frac{x_n}{p_n} + O_{\mathbb{P}}(x_n^2)$$

retains a first-order stochastic term. In the joint limit, the correct order is $O_{\mathbb{P}}\{(Sh^{K/2})^{-1/2}\}$. Moreover, summing pointwise errors over N observations requires lower bounds or tail control for p_n, dependence control, and a triangular-array argument. These are not supplied.

7.5 Lemma 5: convergence of objective values does not control maximisers

An argmax result requires uniform convergence of the criterion and separation or local curvature around the maximiser. A bounded Hessian gives an upper quadratic bound. To infer that two maximisers are close from a small objective difference, one needs a lower quadratic bound or a negative Hessian whose smallest eigenvalue is bounded away from zero after normalisation. Pointwise convergence and identification alone are insufficient.

The structure of the argument makes this explicit. Summing the two second-order expansions leaves a difference of objective values on the left and a term proportional to $(\widehat{\theta}_N - \widehat{\theta}_{N,M,S})^2$ on the right, whose coefficient is an average of two Hessians. Assumption 4 of Brandt and Santa-Clara (2002) bounds that coefficient above. Dividing a small left-hand side by a coefficient that is only bounded above yields no bound at all on the squared distance; what is required is a bound below, that is, a uniformly negative definite Hessian in a neighbourhood. The conclusion drawn, $\widehat{\theta}_{N,M,S} - \widehat{\theta}_N = o(N^{1/2}/S^{1/4})$, also inherits whatever is wrong with the $o(N/S^{1/2})$ input from Lemma 4.

7.6 Lemmas 6 and 7: exact-MLE asymptotics require additional structure

Consistency of the exact diffusion MLE ordinarily requires a uniform law of large numbers, global or local identification, and stability assumptions such as stationarity and ergodicity when $N \to \infty$. A score expansion around θ_0 is not by itself a proof of consistency because the intermediate Hessian is evaluated between the estimator and θ_0.

Similarly, the martingale central limit theorem invoked from Hall and Heyde (1980) requires convergence of the realised conditional quadratic variation and a conditional Lindeberg condition. Defining Fisher information as an expectation does not make the realised conditional variance condition automatic. Boundedness of $\partial p/\partial\theta$ is also not a bound on the score $(\partial p/\partial\theta)/p$ when p can be small; strict positivity of p at each point is weaker than the uniform lower bound the argument needs.

These points do not imply that no correct exact-MLE theorem can be written. They imply that the short proof in Appendix A does not establish the stated result, even apart from the false transition-density joint limit.

7.7 What a counterexample to the argmax estimator would require

It is worth being explicit about the limit of the negative results above. Corollary 3.7 refutes a statement about the simulated transition density at one pair of points. It does not refute the conclusion of Theorem 1, which concerns the maximiser of the simulated log likelihood. Failure of a criterion pointwise does not imply failure of its argmax, and three obstacles separate the two.

First, the collapse is driven by a rare-event geometry that is not uniform over the state space or over the parameter. Corollary 3.7 fixes $x = y = 0$. An argmax argument needs control of $\widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta)$ simultaneously over N observed pairs and over a neighbourhood of θ_0, and the sets on which individual densities collapse need not align.

Second, the simulated criterion is evaluated with common random numbers across θ, as the published implementation requires for a smooth objective. The summands are therefore strongly dependent across parameter values. A criterion that is badly behaved at each fixed θ may still have a well-behaved maximiser if the errors move together; conversely, common random numbers can create a random limit criterion whose argmax is random. Which happens is a property of the coupling, not of the pointwise limit.

Third, the logarithm is unbounded below. Collapse towards zero at a single observation drives one log-density term to $-\infty$, which may push the maximiser away from the region where collapse occurs rather than towards a wrong value. The direction of that effect depends on how the collapse region varies with θ.

A complete argmax counterexample would need to exhibit a parametric family, an admissible sequence (M_n, S_n), and either a limit criterion whose maximiser differs from θ_0, or a demonstration that the simulated criterion fails to converge uniformly on a neighbourhood in a way that the maximiser inherits. We record in Section A the approaches we attempted and where each fails. Until such a construction exists, the correct statement is that the published proofs of Theorems 1 and 2 do not establish their conclusions, not that those conclusions are false. Corollary A.6 shows this is not merely a formal caution: in the location model the maximiser is consistent along the sequences on which the density collapses, so in at least one case the conclusion survives the failure of the proof.

8 Finite-sample scaling diagnostics for a published design

The results so far are asymptotic. This section evaluates them at one concrete design, the implementation of Brandt and Santa-Clara (2002), because a rate statement becomes considerably more informative when a reader can see what it gives at the sample sizes people actually use. The remaining questions about that application, which concern its specification rather than its simulation design, are treated in the companion note of Section 1.2.

Before turning to the structure of the model it is worth recording how the implemented simulation design sits relative to the asymptotic ordering that the published theorems describe. Theorem 2 of Brandt and Santa-Clara (2002) imposes two conditions jointly,

$$\frac{\sqrt{S}}{M} \longrightarrow 0, \qquad \frac{N}{S^{1/4}} \longrightarrow 0, \tag{60}$$

the second entering through the $o(N^{1/2}/S^{1/4})$ bound of their Lemma 5. A sequence satisfying both must eventually have $N^4 \ll S \ll M^2$.

What follows is a set of diagnostics, not a test. A single finite triple (N, M, S) cannot satisfy or violate a limit; only a sequence can. The question a diagnostic can answer is narrower and still useful: are the finite ratios appearing in (60) small, so that the implementation resembles a point along such a sequence, and does the theorem supply any error bound at the implemented sizes? The answer to the second question is no in any case, because (60) is a statement about limits and carries no quantitative finite-sample content.

The estimation uses $N = 544$ weekly observations from January 1990 to May 2000, $M = 10$ Euler substeps, and $S = 5{,}000$ simulations together with 5,000 antithetic variates. Table 4 reports the ratios. At the implemented values $\sqrt{S}/M = 7.07$ and $N/S^{1/4} = 64.7$. Neither is small, so the implementation does not resemble a design drawn from a sequence along which (60) is operating. That is the whole of the claim; nothing here shows that the estimates are inaccurate.

The antithetic variates deserve a separate word, and a careful one. The paired innovation vectors ε and $-\varepsilon$ are deterministic opposites, but the endpoint density contributions are nonlinear functions of them, so their correlation is not -1 and need not even be negative. It depends on the model, the state, the parameter and the endpoint. For the Brownian benchmark it can be computed exactly, and the answer is not the favourable one.

Because S is fixed throughout Section 2 to Section 6 as the number of independent endpoint-density contributions, we do not reuse it here. For this subsection only, let

$P =$ number of independent antithetic pairs, $\quad n_{\text{eval}} = 2P =$ number of density evaluations,

For pair i write G_i^+ and G_i^- for the two contributions and $A_i = \frac{1}{2}(G_i^+ + G_i^-)$ for the pair average, so the antithetic estimator is

$$\widehat{q}_{\text{ant}} = \frac{1}{P}\sum_{i=1}^{P} A_i = \frac{1}{2P}\sum_{i=1}^{P}(G_i^+ + G_i^-). \tag{61}$$

Proposition 8.1 (Antithetic pairing at a coincident endpoint). *For K-dimensional Brownian motion started at x, let G^+ denote the endpoint summand* (8) *generated from a preterminal draw Z and let G^- denote its antithetic partner, generated from the reflected innovations. If $y = x$ then*

$$G^- = G^+ \qquad \textit{almost surely}, \tag{62}$$

so the pair is perfectly positively *correlated and provides no variance reduction whatsoever. More generally, if* $\operatorname{Var}(G_i^\pm) = \sigma^2$ *and* $\operatorname{Corr}(G_i^+, G_i^-) = \rho_A$*, then*

$$\operatorname{Var}(A_i) = \frac{\sigma^2(1+\rho_A)}{2}, \qquad \operatorname{Var}(\widehat{q}_{\text{ant}}) = \frac{\sigma^2(1+\rho_A)}{2P}. \tag{63}$$

At $\rho_A = +1$ this equals σ^2/P, the variance of only P independent evaluations, although $n_{\text{eval}} = 2P$ were computed. At $\rho_A = -1$ it is zero.

Definition 8.2 (Variance-equivalent evaluation count)**.** For an antithetic design of P pairs and *eval* $= 2P$ evaluations with pair correlation $\rho_A \in (-1, 1]$, the *variance-equivalent number of independent evaluations* is the number N_{eff} of independent draws whose average would have the same variance:

$$N_{\text{eff}}(\rho_A) = \frac{n_{\text{eval}}}{1+\rho_A} = \frac{2P}{1+\rho_A}. \tag{64}$$

Two ratios follow from (64), and they must not be confused, because they move in opposite directions:

$$\frac{n_{\text{eval}}}{N_{\text{eff}}} = 1 + \rho_A, \qquad \frac{N_{\text{eff}}}{P} = \frac{2}{1+\rho_A}. \tag{65}$$

The first says how far the raw evaluation count n_{eval} misstates the effective count; the second compares the effective count with the number of independent units. At $\rho_A = +1$ the first is 2 and the second is 1.

The interpretation in the four cases is as follows.

- $\rho_A = +1$: $N_{\text{eff}} = P = n_{\text{eval}}/2$. The *eval* density evaluations have the variance of only P independent evaluations, so counting *eval* as independent overstates the variance-equivalent count by a factor of two.
- $\rho_A = 0$: $N_{\text{eff}} = n_{\text{eval}}$. The antithetic evaluations perform exactly like *eval* independent ones.
- $-1 < \rho_A < 0$: $N_{\text{eff}} > n_{\text{eval}}$. Pairing delivers variance reduction beyond what *eval* independent evaluations would give, which is the case antithetic sampling is designed for.
- $\rho_A \to -1$: N_{eff} diverges, because the idealised pair average has variance tending to zero. This is a statement about a variance-equivalence convention, not a literal count of independent observations; no finite design produces infinitely many of those.

In short, n_{eval} overstates the effective count by the factor $1 + \rho_A$ when $\rho_A > 0$, equals it when $\rho_A = 0$, and understates it when $\rho_A < 0$.

Proof. Reflecting every innovation maps the preterminal state $Z = x + \sqrt{1-h}\,\xi$ to $Z^- = x - \sqrt{1-h}\,\xi$, so $Z^- - x = -(Z - x)$. The summand depends on Z only through $\|y - Z\|^2$, and at $y = x$ this is $\|x - Z\|^2 = \|Z - x\|^2$, which is invariant under $Z - x \mapsto -(Z - x)$. Hence $G^- = G^+$ pointwise, giving (62) and $\rho_A = +1$. For (63), $\operatorname{Var}(G_i^+ + G_i^-) = 2\sigma^2(1 + \rho_A)$, so $\operatorname{Var}(A_i) = \frac{1}{4} \cdot 2\sigma^2(1+\rho_A) = \sigma^2(1+\rho_A)/2$; the pairs are independent across i, so the average of P of them divides this by P. $\square$

Proposition 8.1 is the case $\rho_A = +1$, where the n_{eval} evaluations buy exactly what P would have bought. Numerically, at $h = 1/10$ and $K = 4$, ρ_A is $+1$ at $\|y - x\| = 0$, $+0.063$ at $\|y - x\| = 0.5$ and -0.012 at $\|y - x\| = 1.5$; at $K = 1$ and $\|y - x\| = 0.5$ it is -0.479. These values are illustrative of the mechanism at particular endpoints, not summaries of the estimator's behaviour over a sample.

That last point needs emphasis, because it is what stops N_{eff} from being a single number for a given design. The pair correlation ρ_A depends on the endpoint y, the initial state x, the state dimension K, the discretisation level h, the diffusion coefficients and the parameter value. Every transition in the likelihood therefore carries its own ρ_A, and a single variance-equivalent count for a 544-term log likelihood does not exist without evaluating the pair covariance transition by transition. What can be said without that evaluation is that $\rho_A \leq 1$ always, so $N_{\text{eff}} \geq E/2 = P$, and the coincident-endpoint case attains the bound.

Accordingly we report two distinct scaling diagnostics rather than one effective size:

$$R_{\text{pair}} = \frac{P}{M^{K/2}}, \qquad R_{\text{var}} = \frac{N_{\text{eff}}}{M^{K/2}} = \frac{2P}{(1+\rho_A)M^{K/2}}. \tag{66}$$

R_{pair} counts independent units and requires no knowledge of ρ_A, and it is the conservative diagnostic in the sense that $R_{\text{var}} \geq R_{\text{pair}}$ always. Lemma 8.3 shows that independent pair averages preserve the dimensional consistency scale $P/M^{K/2}$; it does not carry the theory of Section 4 across wholesale. A complete antithetic central limit theorem would additionally require the limiting joint second and third moments of each pair, none of which the lemma supplies, as Remark 8.5 records. R_{var} is the variance-equivalent diagnostic and is known only once ρ_A is. Neither quantity is an exact variance equivalence for the antithetic design taken as a whole: Definition 2.1 is stated for independent contributions, so R_{pair} is its literal instance under the substitution $S \mapsto P$, and R_{var} is a per-transition quantity that varies with ρ_A.

For the reported design, $P = 5{,}000$ independent pairs and $n_{\text{eval}} = 10{,}000$ endpoint-density evaluations. Table 4 reports both, with R_{pair} and with R_{var} evaluated at two reference values of ρ_A rather than at a single assumed one.

Finally, none of this touches the dimensional exponent. Pairing changes the variance constant and can change it in either direction, but the $h^{-K/2}$ growth of the second moment survives it.

Lemma 8.3 (Antithetic pairing preserves the endpoint moment exponent)**.** *Let G_h^+ and G_h^- be nonnegative with identical marginal laws and let $A_h = \frac{1}{2}(G_h^+ + G_h^-)$, with no assumption on their dependence. Then*

$$\tfrac{1}{2}\,\mathbb{E}[G_h^2] \;\leq\; \mathbb{E}[A_h^2] \;\leq\; \mathbb{E}[G_h^2], \tag{67}$$

where G_h denotes the common marginal. Consequently, if $\mathbb{E}[G_h^2] \asymp h^{-K/2}$ then $\mathbb{E}[A_h^2] \asymp h^{-K/2}$; and if in addition $\mathbb{E}[A_h] = O(1)$, then $\mathrm{Var}(A_h) \asymp h^{-K/2}$.

Proof. Write $a = G_h^+$ and $b = G_h^-$, both nonnegative. Then $4A_h^2 = (a+b)^2 = a^2 + b^2 + 2ab$. Since $ab \geq 0$, we get $4A_h^2 \geq a^2 + b^2$; and since $2ab \leq a^2 + b^2$, we get $4A_h^2 \leq 2(a^2 + b^2)$. Hence

$$\frac{a^2+b^2}{4} \;\leq\; A_h^2 \;\leq\; \frac{a^2+b^2}{2},$$

where nonnegativity is used only for the lower bound. Taking expectations and using $\mathbb{E}[a^2] = \mathbb{E}[b^2] = \mathbb{E}[G_h^2]$ gives (67). Both bounds are constants free of h and of the dependence structure, so the two second moments have the same order, and $\mathbb{E}[A_h^2] \asymp h^{-K/2}$ follows from $\mathbb{E}[G_h^2] \asymp h^{-K/2}$.

For the variance statement, $\mathrm{Var}(A_h) = \mathbb{E}[A_h^2] - \mathbb{E}[A_h]^2$. The subtracted term is $O(1)$ by hypothesis while $\mathbb{E}[A_h^2]$ diverges at rate $h^{-K/2}$, so it is negligible and $\mathrm{Var}(A_h) \asymp h^{-K/2}$. The hypothesis $\mathbb{E}[A_h] = O(1)$ holds in the present setting because $\mathbb{E}[A_h] = \mathbb{E}[G_h]$, which converges to the transition density. □

So the dimensional law of Theorem 4.2 is untouched by the coupling: only the constant in front of $h^{-K/2}$ responds to ρ_A, and Lemma 8.3 bounds that response within a factor of two of the independent case.

Nonnegativity says something sharper about ρ_A itself, and it is not encouraging for the design.

Proposition 8.4 (Antithetic pairing cannot help at leading order)**.** *Suppose $G_h^+, G_h^- \geq 0$ have the common marginal law of G_h, with $\mathbb{E}[G_h] = O(1)$ and $\mathrm{Var}(G_h) \asymp h^{-K/2}$. Then*

$$\rho_A \;\geq\; -\frac{\mathbb{E}[G_h]^2}{\mathrm{Var}(G_h)} \;=\; O(h^{K/2}), \tag{68}$$

so $\liminf_{h\downarrow 0}\rho_A \geq 0$*. Consequently* $N_{\mathrm{eff}} \leq n_{\mathrm{eval}}(1+o(1))$*: relative to* n_{eval} *independent evaluations, antithetic pairing cannot* reduce *the leading rare-event variance constant as the mesh is refined.*

Proof. Since both variables are nonnegative, $\mathbb{E}[G_h^+ G_h^-] \geq 0$, so

$$\mathrm{Cov}(G_h^+, G_h^-) = \mathbb{E}[G_h^+ G_h^-] - \mathbb{E}[G_h]^2 \;\geq\; -\mathbb{E}[G_h]^2.$$

Dividing by $\mathrm{Var}(G_h)$ gives (68). The numerator is $O(1)$ by hypothesis and the denominator diverges at rate $h^{-K/2}$, so the bound tends to zero and $\liminf \rho_A \geq 0$. The statement about $N_{\mathrm{eff}} = n_{\mathrm{eval}}/(1+\rho_A)$ follows immediately. □

The mechanism is worth naming. A negative ρ_A requires the pair to be jointly small when one member is large, but both members are nonnegative densities whose variance is driven by a rare event in which one of them is enormous; on that event the product is nonnegative, and the covariance cannot be more negative than $-\mathbb{E}[G_h]^2$, which does not grow.

The direction of the conclusion should be read carefully, because Proposition 8.4 is one-sided. It says antithetic pairing cannot reduce the leading $h^{-K/2}$ variance constant below the independent-evaluation constant. It does not say the constant is unchanged: if $\rho_A \to c > 0$, the constant is multiplied by $1+c$, and at a coincident endpoint $\rho_A = 1$ doubles it relative to n_{eval} independent evaluations. So the possibilities the proposition leaves open are that pairing leaves the leading constant alone or makes it worse, never that it improves it. Antithetic sampling can still reduce variance at a *fixed h* and a favourable endpoint, as the value $\rho_A = -0.479$ above shows, and that is a real finite-sample gain. In the theoretical sections S continues to mean the number of independent contributions.

Remark 8.5 (What Lemma 8.3 does not give)**.** The lemma fixes the order of the second moment and, with $\mathbb{E}[A_h] = O(1)$, of the variance. It does not give the exact local variance constant, the third centred-moment constant, the Lyapunov ratio for a pair-level triangular array, or a central limit theorem at either the pair level or the likelihood level. Establishing an antithetic analogue of Theorem 4.13 would require the limiting joint second and third moments of (G_h^+, G_h^-), which depend on the coupling and are not supplied by a sandwich bound.

The dimensional diagnostic of Definition 2.1 gives the complementary reading. With $K = 4$, $M = 10$ and $P = 5{,}000$ independent pairs,

$$R_{\mathrm{pair}} = \frac{P}{M^{K/2}} = \frac{5{,}000}{100} = 50. \tag{69}$$

Each transition density in a 544-term log likelihood is therefore resolved by an independent-pair count of order 10^2, not the 10^4 that the raw evaluation total suggests. The variance-equivalent figure R_{var} lies between 50 and 100 wherever $\rho_A \geq 0$ and above 100 wherever $\rho_A < 0$, so the order of magnitude is 10^2 on any reading short of near-perfect negative pairing. A scaling statement invites the question it does not answer, namely whether 50 is adequate, and Proposition 3.1 answers it for the Brownian benchmark rather than leaving the reader to wonder. At a coincident endpoint the exact second moment gives

$$\frac{\mathbb{E}[G_h^2]}{p(y \mid x)^2} = \big(h(2-h)\big)^{-K/2} = 27.70 \qquad \text{at } K = 4,\ h = \tfrac{1}{10}, \tag{70}$$

so the relative standard deviation of the simulated density from $S = 5{,}000$ independent units is

$$\sqrt{\frac{27.70 - 1}{5{,}000}} = 7.3 \text{ per cent},$$

Table 4: Finite-sample scaling ratios for the implemented design and three variations, at $N = 544$ and $K = 4$. Here P is the number of antithetic pairs and $n_{\text{eval}} = 2P$ the number of endpoint-density evaluations. The identification $S = P$ needs a word, because S was defined in Section 2 as the number of independent endpoint contributions and an antithetic design produces $2P$ contributions, not P. They are not independent: the two members of a pair are deterministically linked, so the independent units are the P *pair averages*, and it is those that play the role S plays in the theory. Substituting $S = P$ is therefore the correct transfer, and it is conservative, since by Proposition 8.4 a pair average is never worth more than two independent evaluations asymptotically and is worth exactly one at a coincident endpoint. $R_{\text{pair}} = P/M^2$ counts independent units and needs no knowledge of ρ_A; it is Definition 2.1 under the substitution $S \mapsto P$, and is a conservative diagnostic rather than an exact variance equivalence. $R_{\text{var}} = N_{\text{eff}}/M^2$ with $N_{\text{eff}} = n_{\text{eval}}/(1 + \rho_A)$ is the variance-equivalent diagnostic, and it is reported at *two reference values* of ρ_A because no single value applies to a 544-term likelihood: $\rho_A = 1$ is the worst case, attained at a coincident endpoint, and $\rho_A = 0$ the neutral case in which pairing neither helps nor hurts. No separate N_{eff} column is needed at these two values, because $N_{\text{eff}}(1) = P$ and $N_{\text{eff}}(0) = n_{\text{eval}}$: the variance-equivalent count is read off the P and n_{eval} columns already shown. Any $\rho_A < 0$ gives R_{var} larger than both columns shown, and $R_{\text{var}} \geq R_{\text{pair}}$ always. These are diagnostics: the columns record how the implementation is placed, not whether any limit holds.

					$R_{\text{var}} = N_{\text{eff}}/M^2$			
Scenario	M	$P = S$	$n_{\text{eval}} = 2P$	R_{pair}	$\rho_A = 1$	$\rho_A = 0$	$\sqrt{P}/M$	$N/P^{1/4}$
Implemented	10	5,000	10,000	50.0	50.0	100.0	7.07	64.7
Double M and P	20	10,000	20,000	25.0	25.0	50.0	5.00	54.4
Double M, P fixed	20	5,000	10,000	12.5	12.5	25.0	3.54	64.7
Double M, quadruple P	20	20,000	40,000	50.0	50.0	100.0	7.07	45.7

rising to 9.3 per cent at $\|y - x\| = 1$, where the ratio in (70) is 44.48.

Two readings of that figure are both correct and should be kept apart. Pointwise, seven per cent on a single transition density is not catastrophic. In the criterion it is different: by the second-order term of Proposition 6.1 the log-density expectation is displaced by approximately $-\operatorname{Var}(\widehat{q})/(2q^2)$ per observation, which at these values is -2.7×10^{-3}, and summing over the 544 observations gives a systematic displacement of about -1.5 in log-likelihood units, rising to -2.4 at the more distant endpoint. A bias of that size does not move a maximiser much when it is common across parameter values, but it does not cancel in comparisons *across specifications* with different fitted densities, which is exactly the use to which a log likelihood is put when models are ranked. That is the practical content of $R_{M,S} = 50$, and it is available only because the moments are exact rather than asymptotic.

The last three rows of Table 4 record how $R_{M,S}$ responds to refinement, because the arithmetic is easy to get wrong. In four dimensions $R_{M,S} = S/M^2$, so

$$R_{2M,2S} = \frac{2S}{4M^2} = \frac{R_{M,S}}{2}, \qquad R_{2M,S} = \frac{R_{M,S}}{4}, \qquad R_{2M,4S} = R_{M,S}. \tag{71}$$

The paper reports a final round of optimisation in which both the simulation count and M are doubled. In four dimensions that halves the independent-pair scaling diagnostic $R_{\text{pair}} = P/M^2$ rather than increasing it. The corresponding change in the variance-equivalent diagnostic R_{var} cannot be determined without the antithetic pair correlation at each transition, since ρ_A may itself vary with M; what can be said is that $R_{\text{var}} \geq R_{\text{pair}}$ throughout. The reported stability of the estimates under this refinement is therefore not evidence that the simulation is converging in the endpoint dimension; it is a comparison between two designs, the second of which has half the independent-pair scaling diagnostic of the first. Its variance-equivalent size may change by a different factor, because the pair correlation may change with M. Holding R_{pair} fixed while doubling M requires quadrupling the number of pairs, which is the last row of the table.

9 Numerical evidence

All numerical results in this section are generated by the accompanying Python script. They use exact Gaussian benchmarks where possible and are intended to illustrate the theorems rather than replace them.

9.1 Second-moment scaling

Figure 1 plots

$$h^{K/2}\mathbb{E}[G_h^2]$$

divided by its limiting value for $K \in \{1, 2, 4, 8\}$. The exact formula (11) converges to the constant predicted by Theorem 4.2. The unscaled second moment grows as $M^{K/2}$.

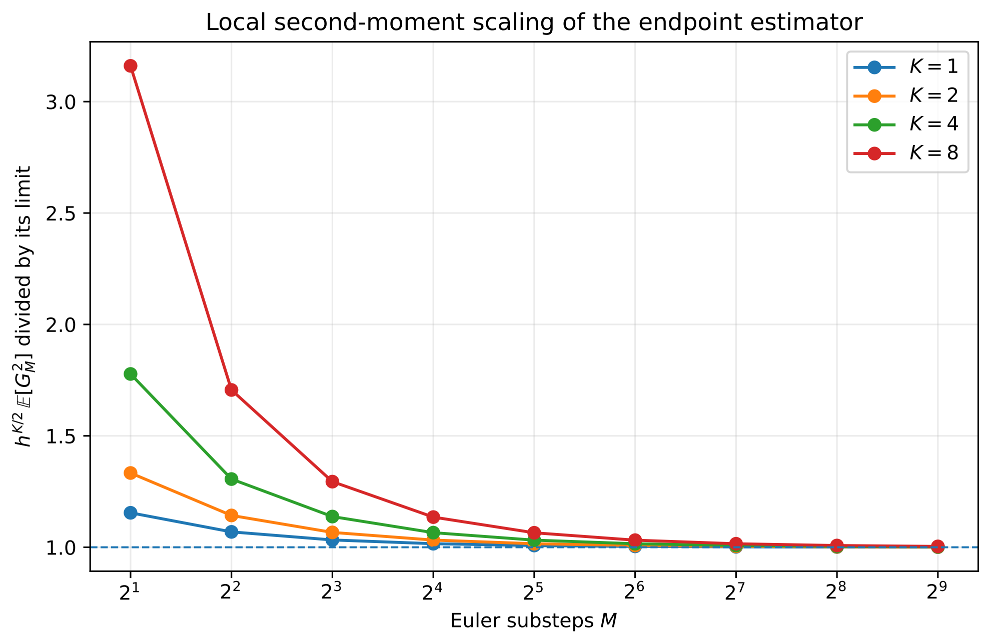

Figure 1: Exact local second-moment scaling for standard Brownian motion. The plotted quantity converges to one after division by its theoretical limit.

9.2 Three joint asymptotic paths

For $K = 4$, Figure 2 compares the exact relative root mean squared error under $S = M$, $S = M^2$, and $S = M^3$. The first path satisfies the published condition but has increasing RMSE. The second keeps the effective endpoint size constant and does not become consistent. The third satisfies $S/M^2 \to \infty$ and has decreasing RMSE.

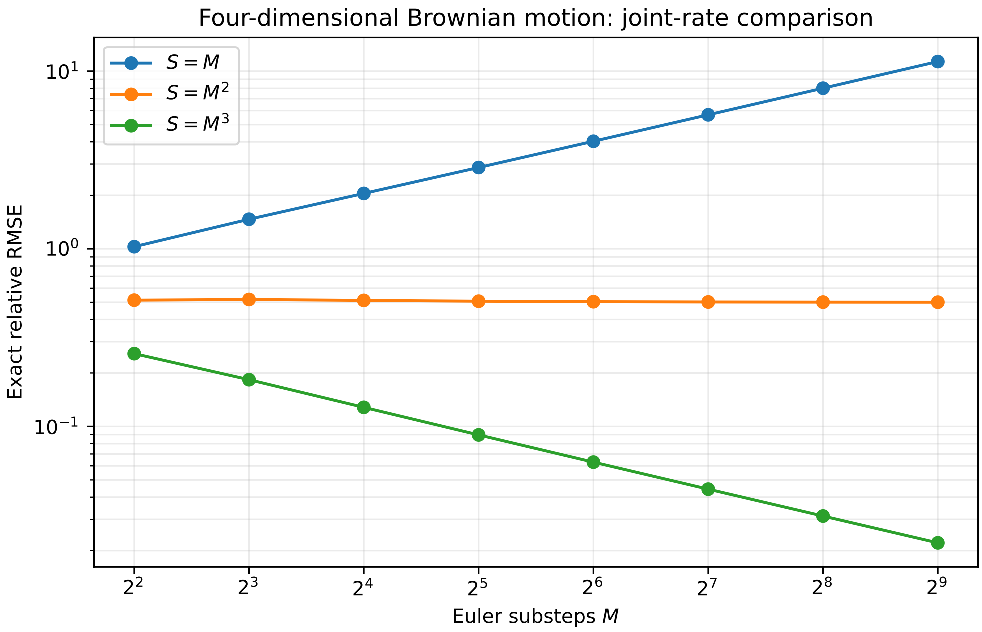

Figure 2: Exact relative RMSE at the origin for four-dimensional Brownian motion.

9.3 Typical estimates collapse while the mean remains correct

Figure 3 simulates the counterexample path $M = S$. The sample median falls rapidly towards zero, and by $M = 512$ more than 95% of estimates are below half the true density. The sample mean remains near one after normalisation because a small number of very large realisations carry the expectation.

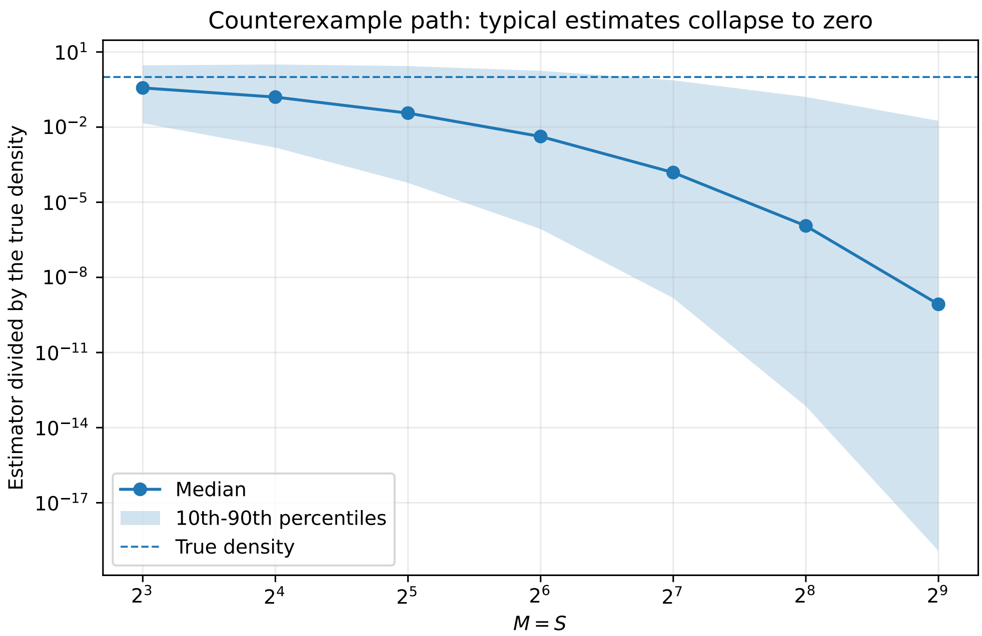

Figure 3: Monte Carlo distribution along $M = S$ for four-dimensional Brownian motion. Values are divided by the true density. The dashed line is one.

Table 5: Simulation of the four-dimensional Brownian counterexample path $M = S$. The estimator is divided by the true density. Every row uses 20,000 replications, so the quantiles are estimated with equal precision across the table; an earlier design held the total draw count fixed instead, which estimated them worst at the largest M, precisely where the collapse is sharpest. The parenthesised figure is the Monte Carlo standard error of the mean. That column needs no draws in principle, since Proposition 3.1 makes the estimator exactly unbiased at every M and the population mean is 1; it is retained because the contrast between a mean pinned at one and a median falling through ten orders of magnitude is the content of the table. Its standard error grows from 0.010 to 0.073 across the rows, so the entry 0.879 at $M = 512$ sits within 1.7 standard errors of unity and reflects the heaviness of the tail rather than any failure of unbiasedness. The median and lower quantiles are where the collapse of Corollary 3.7 is visible.

M	Replications	Mean	(s.e.)	Median	10th pct.	90th pct.	$\mathbb{P}(\widehat{q} < 0.5p)$
8	20000	1.001	(0.010)	0.3674	1.427×10^{-2}	2.940	0.560
16	20000	1.009	(0.015)	0.1582	1.520×10^{-3}	3.161	0.665
32	20000	1.027	(0.021)	0.03628	5.925×10^{-5}	2.746	0.754
64	20000	1.002	(0.028)	0.004223	8.507×10^{-7}	1.777	0.827
128	20000	1.042	(0.042)	1.537×10^{-4}	1.487×10^{-9}	0.7409	0.886
256	20000	0.953	(0.054)	1.138×10^{-6}	7.158×10^{-14}	0.1607	0.929
512	20000	0.879	(0.073)	8.409×10^{-10}	1.193×10^{-19}	0.01790	0.954

9.4 A nonzero drift does not remove the dimensional law

Consider the diagonal Ornstein–Uhlenbeck process

$$dY_t = -Y_t dt + dW_t$$

in four dimensions. Its exact transition density is known, and the preterminal Euler state is Gaussian, so the second moment of the endpoint estimator can again be computed exactly. Figure 4 shows convergence to the local constant of Theorem 4.2. The phenomenon is therefore not specific to driftless Brownian motion.

One point of bookkeeping should be stated rather than glossed. The drift $\mu(y) = -y$ is *not* bounded, so this process does not satisfy (A1) and is not covered by Theorem 4.2 as that theorem is stated. The agreement in Figure 4 is therefore evidence that the conclusion extends beyond the hypotheses under which it is proved, not a verification of the theorem on one of its own instances. It is the localised form sketched in Remark 4.5, which needs ellipticity and drift boundedness only on a ball around y together with a Gaussian upper bound on the preterminal density, that would cover an Ornstein–Uhlenbeck process; we have not written that version out, and until it is written the figure should be read as an illustration rather than as a confirmation.

10 Remedies and alternative constructions

The endpoint variance problem is not an unavoidable property of likelihood inference for diffusions. It is a consequence of drawing preterminal states from the unconditioned forward Euler law and asking a shrinking final Gaussian kernel to connect them to a fixed observed endpoint.

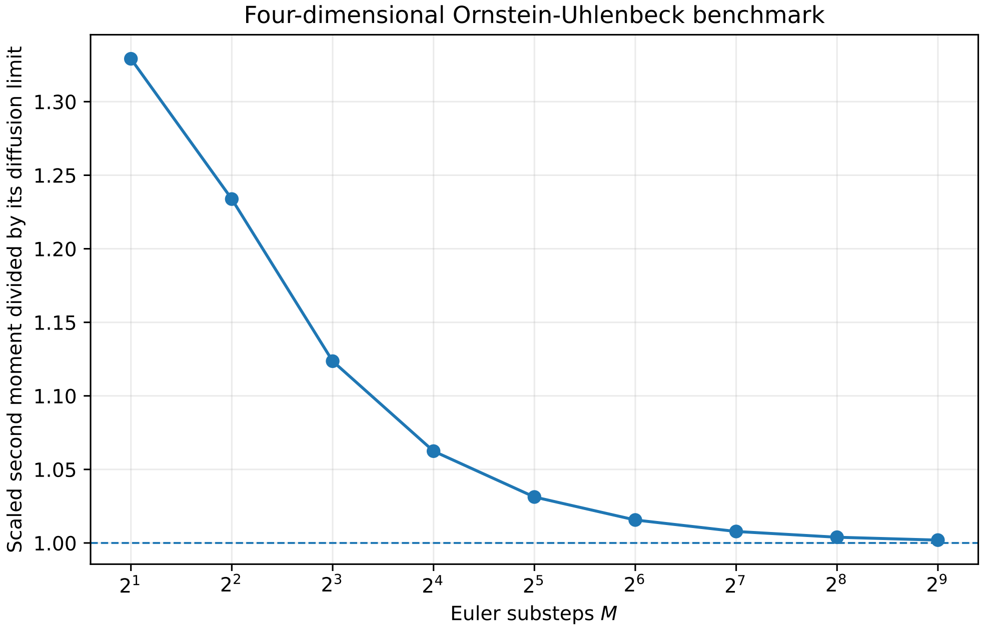

Figure 4: Scaled second moment for a four-dimensional Ornstein–Uhlenbeck diffusion.

10.1 Bridge and importance proposals

A bridge proposal guides simulated paths towards the observed endpoint. Bladt and Sørensen (2014) give a construction requiring no auxiliary importance weights, which is the cleanest available answer to the mechanism identified here. In importance-sampling language, it increases the probability of drawing paths in the region that contributes to the transition-density integral and corrects the proposal through a likelihood ratio. This is the direction taken by the acceleration methods studied by Durham and Gallant (2002), the path-augmentation schemes of Elerian et al. (2001), conditioned-diffusion methods such as Delyon and Hu (2006), the regularised proposal of Lindström (2012), and later guided proposals for multidimensional bridges (Schauer et al., 2017; Meulen and Schauer, 2017). The nonparametric simulation approach of Nicolau (2002) replaces the endpoint Gaussian kernel by a density estimate, which changes the bandwidth problem rather than removing it.

A useful proposal should be assessed through the moments of its importance weights, and this is the respect in which bridge proposals differ in kind rather than in degree. Stramer and Yan (2007a) show that the modified Brownian bridge has importance-weight variance bounded uniformly in M, which is exactly what fails here: Corollary 4.10 gives $\mathrm{Var}(G_h) \asymp h^{-K/2}$ for the endpoint summand. Their numerical comparison makes the contrast visible, the endpoint estimator's error increasing with M at fixed N while the bridge's does not, and their accuracy for the bridge is dimension-free where the endpoint rate is not. Conditioning on the endpoint therefore addresses the rare-event geometry identified here at its source, rather than merely reducing a constant.

10.2 Exact or discretisation-free likelihood estimators

For restricted classes of diffusions, exact simulation and unbiased transition-density estimators remove Euler bias entirely. Beskos, Papaspiliopoulos, Roberts, and Fearnhead (2006) and Beskos, Papaspiliopoulos, and Roberts (2009) develop likelihood-based methods of this kind. Their applicability is model dependent, but they illustrate the correct theoretical structure:

unbiasedness, continuity in parameters, finite-moment conditions, and joint control of data and Monte Carlo sample sizes are stated explicitly.

10.3 Analytical density expansions

Closed-form expansions avoid endpoint Monte Carlo variance entirely but introduce an approximation-order problem of their own. For the four-dimensional system at issue here the relevant reference is the multivariate expansion of Aït-Sahalia (2008) rather than the univariate one of Aït-Sahalia (2002): it constructs the expansion for a multivariate diffusion after a reducibility transformation, and it is the natural competitor to a simulated likelihood at $K = 4$. Its applicability turns on whether the model can be transformed to satisfy the required regularity conditions, which for square-root states with a Feller ratio of one half is not automatic. Approximate martingale estimating functions, as in Kessler (1997), are a further discretisation-bias-free alternative that avoids the transition density altogether at some cost in efficiency.

10.4 Positivity-preserving discretisation

For square-root components, ordinary Euler should be replaced by an exact transition where available or by a specified positivity-preserving approximation. The choice is part of the estimator, not an implementation detail, because different truncation and reflection rules define different transition densities.

This is a solved problem for the square-root process, which is why leaving the rule unspecified is a real omission rather than a pedantic one. Alfonsi (2005) classifies the discretisation schemes for the Cox–Ingersoll–Ross and squared-Bessel processes and characterises which preserve positivity, and Alfonsi (2010) constructs higher-order schemes with the same property. Higham et al. (2002) give the general strong-convergence theory for Euler-type methods when the coefficients fail the usual global Lipschitz and linear-growth conditions, which is exactly the regime a square root creates at the origin. For the incompleteness state of the exchange-rate model, the companion note observes a simpler route still: the natural coordinate is the Ornstein–Uhlenbeck process whose square it is, and there the transition is exactly Gaussian and no positivity rule is needed at all.

10.5 Practical design rule

If the unconditioned endpoint estimator is nevertheless used, the practical content of this paper reduces to four points, which we gather here because they are otherwise distributed across the results above.

One: monitor the right quantity. The diagnostic is not S but the effective endpoint size

$$R_{M,S} = \frac{S}{M^{K/2}},$$

and for parameter estimation rather than density evaluation it is the score-level version $S/M^{(K+2)/2}$ of (58), which is more demanding by one power of M. Increasing M without increasing these is not an accuracy improvement.

Two: refining both at once makes things worse, not better. This is the point most likely to be missed in practice, because refining M and S together looks like unambiguous progress. In $K = 4$,

$$R_{2M,2S} = \frac{2S}{(2M)^2} = \frac{R_{M,S}}{2}, \tag{72}$$

so doubling the substeps and the simulation count halves the effective size. Holding $R_{M,S}$ fixed while doubling M requires *quadrupling* S; and in K dimensions, holding it fixed requires $S \propto M^{K/2}$. Stability of estimates under a joint refinement is therefore not evidence of convergence, since the refinement has moved the design in the unfavourable direction.

Three: know where the implemented design sits. At the values used by Brandt and Santa-Clara (2002), $K = 4$, $M = 10$ and 5,000 independent simulation units, $R_{M,S} = 50$. Each transition density in a 544-term log likelihood is resolved by an effective sample of order 10^2, not 10^4. That may be adequate; the point is that it is the number to check, and it is not the one the design appears to supply.

Four: prefer a proposal that uses the endpoint. The dimensional penalty comes from the unconditioned proposal, and bridge or guided proposals remove it rather than mitigating it: Stramer and Yan (2007a) show the modified Brownian bridge has importance-weight variance bounded uniformly in M, with an optimal allocation that is dimension free. If the endpoint construction is retained, monitor the variance and lower tail of every estimated transition density, use log-sum-exp evaluation, and verify stability along a path that holds $R_{M,S}$ fixed or increasing rather than along $M = S$.

11 Conclusion

The endpoint simulated-likelihood estimator has a simple but consequential dimensional structure. Its final Gaussian transition is a shrinking kernel. In K dimensions, the second moment of one simulated contribution grows as $M^{K/2}$, so the Monte Carlo variance is of order $M^{K/2}/S$. The relevant simulation size is $S/M^{K/2}$.

This leads to four conclusions about Brandt and Santa-Clara (2002), stated in decreasing order of strength. First, two published statements are directly false: the joint transition-density consistency of Lemma 2 and the joint central-limit statement of Lemma 3. An exact Brownian sequence satisfying the paper's own Assumptions 1 and 2, and satisfying its rate condition, makes the simulated density converge in probability to zero rather than to the positive true density, and makes $\sqrt{S}$ times the centred density diverge to $-\infty$. Second, because Lemmas 4 to 6 take those results as inputs, the published proofs of Theorems 1 and 2 fail. That is a statement about the proofs; we do not show that their conclusions are false. Section A goes further and shows, in a Gaussian location model, that the maximiser is consistent along the very sequences on which the density collapses, so the counterexample demonstrably does not transfer. Third, the rate condition used in those theorems is incompatible with pointwise mean-square consistency of the simulated transition density in the four-dimensional application. Fourth, and independently of all of the above, the ratios corresponding to the paper's own conditions $\sqrt{S}/M \to 0$ and $N/S^{1/4} \to 0$ are not small at the implemented design $N = 544$, $M = 10$, $S = 5{,}000$, so that design is not in a regime resembling the asymptotic ordering the theorems describe.

The corrected transition-density analysis also supplies constructive results. The density estimator has a triangular-array CLT under $S/M^{K/2} \to \infty$. If the Euler density bias is first order, the MSE-optimal discretisation is $M \asymp S^{2/(K+4)}$. The nonlinear log-likelihood scale is $M^{K/2}/S$, subject to the usual lower-tail conditions required for expectations of logarithms. These results explain why bridge proposals and other importance-sampling methods are not merely computational refinements: they address the rare-event geometry of the estimator.

The exchange-rate application raises separate questions, which the companion note addresses and which we summarise here only to record their bearing on the above. Its square-root states fall outside the smooth uniformly elliptic theory of Section 4, so the generic theorem does not apply to it as written; the incompleteness state has Feller ratio exactly one half for every admissible parameter value, so zero is attainable and no estimate can move it away from that boundary; the estimator substitutes that state out and simulates volatility instead, so the operative requirement is that a recovered square root stay real along every simulated Euler path, and no convention for handling a violation is documented. The Brownian correlation matrix is positive definite everywhere we could check, and what is missing is only a parameterisation guaranteeing that globally. The final empirical estimator also contains a minimum-incompleteness corner rule and constraints the original authors report as binding, which are not covered by ordinary interior SML asymptotics. None of this is shown to make the implemented chain fail, and several of the findings are negative in exactly that sense; the companion states each with its own scope.

None of these points, individually or collectively, proves that the finite-sample economic conclusions are false. They show that those conclusions cannot inherit the claimed maximum-likelihood asymptotics from the mathematical argument given. The appropriate post-publication response is a corrected theorem, a clear statement of the estimator actually implemented, and a re-examination of numerical stability under dimensionally valid simulation rates or bridge-based alternatives.

A The argmax question in a location model

This appendix records what can be proved about the simulated-likelihood maximiser in the most favourable possible model, and where the argument stops. The conclusion is negative: we do not obtain a counterexample to argmax consistency, and the manuscript makes no such claim. Two intermediate results are complete and are stated here because they clarify what the endpoint construction does to the estimation problem.

Consider

$$dY_t = \theta\, dt + dW_t, \qquad Y_t \in \mathbb{R}^K, \tag{73}$$

observed at unit intervals $t = 0, 1, \ldots, N$. The Euler scheme is exact for every M, the coefficients satisfy Assumptions 1 and 2 of Brandt and Santa-Clara (2002) exactly, and the exact maximum likelihood estimator is the mean increment $\widehat{\theta}_N = N^{-1} \sum_n \Delta Y_n$. Common random numbers across θ are used throughout, as the published implementation requires for a smooth objective: the innovations $\xi_{n,s} \sim \mathcal{N}(0, I_K)$ are drawn once and held fixed as θ varies.

Proposition A.1 (Common random numbers give an exact kernel estimator)**.** *Set $h = 1/M$, $u_n(\theta) = \Delta Y_n - \theta$ and $a_{n,s} = \sqrt{1-h}\,\xi_{n,s}$. Then for every θ,*

$$\widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta) = \frac{1}{S} \sum_{s=1}^{S} \varphi_K(u_n(\theta) - a_{n,s}; 0, hI_K)\,. \tag{74}$$

That is, the simulated transition density is exactly a Gaussian kernel density estimate with bandwidth $\sqrt{h}$, built from a sample that does not depend on θ and evaluated at the θ-dependent point $u_n(\theta)$.

Proof. The Euler recursion for (73) is $Y^M_{m+1} = Y^M_m + \theta h + \sqrt{h}\,\varepsilon_{m+1}$, so after $M-1$ steps from $Y^M_0 = Y_n$ the preterminal state is $Z_{n,s}(\theta) = Y_n + \theta(1-h) + \sqrt{1-h}\,\xi_{n,s}$, where $\sqrt{1-h}\,\xi_{n,s}$ collects the $M-1$ independent increments. The one-step density (3) evaluated at Y_{n+1} has mean $Z_{n,s}(\theta) + \theta h$ and covariance hI_K, and

$$Y_{n+1} - Z_{n,s}(\theta) - \theta h = \Delta Y_n - \theta - \sqrt{1-h}\,\xi_{n,s} = u_n(\theta) - a_{n,s}.$$

Substituting into (5) gives (74). □

Proposition A.1 is an identity, not an approximation, and it recasts Corollary 3.7 in familiar terms: a kernel density estimate with S atoms and bandwidth $\sqrt{h}$ is uninformative when $Sh^{K/2} \to 0$.

Proposition A.2 (Uniform comparison with a nearest-atom objective)**.** *Write* $b_{n,s} = \Delta Y_n - \sqrt{1-h}\,\xi_{n,s}$ *and*

$$D_N(\theta) = \sum_{n=0}^{N-1} \min_{s\le S} \|\theta - b_{n,s}\|^2.$$

Then for every θ*,*

$$-2Nh\log S \le 2h\log\widehat{\mathcal{L}}_{N,M,S}(\theta) + D_N(\theta) + NhK\log(2\pi h) \le 0. \tag{75}$$

Proof. Fix n and let $d_n = \min_s\|\theta - b_{n,s}\|^2$. Factoring the minimising term out of the sum in (74),

$$\frac{1}{S}\sum_s e^{-\|\theta-b_{n,s}\|^2/(2h)} = e^{-d_n/(2h)}\cdot\frac{1}{S}\sum_s e^{-(\|\theta-b_{n,s}\|^2-d_n)/(2h)}.$$

Every exponent in the second factor is nonpositive and at least one is zero, so that factor lies in $[1/S, 1]$ and its logarithm lies in $[-\log S, 0]$. Taking logarithms, adding the constant $-\frac{K}{2}\log(2\pi h)$ from the Gaussian prefactor, multiplying by $2h$ and summing over n gives (75). □

The bound is uniform in θ with slack $2Nh\log S$. After division by N that slack is $2h\log S$, which along $M = S = n$ equals $2(\log n)/n$ and therefore vanishes. What this does *not* settle is whether the comparison is sharp enough to control the maximiser, because that is a question about the slack relative to the variation of D_N near its optimum, and the order and local geometry of D_N are exactly what remain unresolved.

So the statement is the one in the title: the scaled simulated log likelihood lies uniformly within $2Nh\log S$ of a transformed nearest-atom criterion. The simulated criterion is not itself a nearest-neighbour functional; it is a smooth log-sum-exp of NS terms, and Proposition A.2 says only that after scaling and the addition of a constant it is uniformly approximated by one. The exact log likelihood in this model is by contrast the quadratic $-\frac{1}{2}\sum_n\|\theta - \Delta Y_n\|^2$ with minimiser $\widehat{\theta}_N$. Whether the approximation controls the maximiser depends on the unresolved local scale of D_N, so any argument that the two maximisers agree must be made rather than assumed.

Five things the uniform bound alone does not establish, and which are not claimed anywhere in this paper: convergence of the two maximisers; equality of their finite-sample values; equivalence of local curvature at the optimum; the order of D_N; or a contradiction to Lemma 5 of Brandt and Santa-Clara (2002).

What is missing is the limit of D_N, and it is worth being precise about how much is missing.

The candidate limit is computable in closed form. By Proposition A.1 the atoms $b_{n,s} = \Delta Y_n - \sqrt{1-h}\,\xi_{n,s}$ do not depend on θ at all; given ΔY_n they are independent across s with the Gaussian density f_n, and they are independent across n. Classical nearest-neighbour asymptotics for an independent sample from a fixed density then give

$$\mathbb{E}\Big[\min_{s\le S}\|\theta - b_{n,s}\|^2 \;\Big|\; \Delta Y_n\Big] \;\sim\; \Gamma(1+\tfrac{2}{K})\,(\omega_K S)^{-2/K} f_n(\theta)^{-2/K}, \tag{76}$$

with ω_K the volume of the unit ball. Since $f_n(\theta)^{-2/K} = 2\pi(1-h)\exp\{\|\theta-\Delta Y_n\|^2/(K(1-h))\}$ and $\Delta Y_n \sim \mathcal{N}(\theta_0, I)$ in this model, averaging over ΔY_n with the Gaussian identity $\mathbb{E}e^{a\|W\|^2} = (1-2a)^{-K/2}e^{a\|m\|^2/(1-2a)}$ at $a = 1/K$ gives the candidate

$$N^{-1}S^{2/K}D_N(\theta) \;\longrightarrow\; 2\pi\,\Gamma(1+\tfrac{2}{K})\,\omega_K^{-2/K}\Big(\frac{K}{K-2}\Big)^{K/2}\exp\Big(\frac{\|\theta-\theta_0\|^2}{K-2}\Big), \tag{77}$$

finite exactly when $a < 1/2$, that is when $K > 2$. The threshold is not a coincidence of two calculations: it is the integrability boundary of the same Gaussian moment that drives Corollary 3.7, appearing here as the condition for the averaged nearest-neighbour distance to be finite.

The interchange required to pass from (76) to (77) is uniform integrability of $S^{2/K}\min_s\|\theta - b_{n,s}\|^2$ over the distribution of ΔY_n, and it is a genuine obstacle rather than a formality: the limit is dominated by centres far from θ, where (76) needs the largest S, since its accuracy is governed by $Sf_n(\theta)$ and $f_n(\theta)$ is exponentially small there. At $K = 4$ the nearest 90 per cent of centres contribute only 58 per cent of (77), and the outermost 1 per cent contributes 15 per cent. That is why direct simulation settles nothing here: at $K = 4$ and $S = 5{,}000$ it reaches only about 89 per cent of the limit.

The obstacle can nevertheless be removed, by splitting the centres at a radius that grows with S and treating the two regions by different arguments.

Proposition A.3 (The interchange holds)**.** *Fix θ, write $\sigma^2 = 1 - h$ and $m = \|\theta - \Delta Y_n\|$, and suppose*

$$K\sigma^2 > 2. \tag{78}$$

Then, as $S \to \infty$,

$$S^{2/K}\,\mathbb{E}\Big[\min_{s\le S}\|\theta - b_{n,s}\|^2\Big] \longrightarrow \Gamma\big(1+\tfrac{2}{K}\big)\,\omega_K^{-2/K}\,\mathbb{E}\big[f_n(\theta)^{-2/K}\big], \tag{79}$$

the expectation on the right being finite, and the family $\{S^{2/K}\min_s\|\theta - b_{n,s}\|^2\}_{S\ge 1}$ is uniformly integrable. Evaluating the right-hand side by the Gaussian moment identity gives (77)*, which is therefore a theorem and not a conjecture.*

Proof. Write $\rho_S = \min_{s\le S}\|\theta - b_{n,s}\|$ and condition throughout on ΔY_n, so that the atoms are independent with the $\mathcal{N}(\Delta Y_n, \sigma^2 I_K)$ density f_n. Let $F(t) = \mathbb{P}(\|\theta - b_{n,1}\| \le t \mid \Delta Y_n)$, so that

$$\mathbb{E}[\rho_S^2 \mid \Delta Y_n] = \int_0^\infty \big(1 - F(\sqrt{v})\big)^S\,dv \le \int_0^\infty e^{-SF(\sqrt{v})}\,dv. \tag{80}$$

Fix a with $2/K < a < \sigma^2$, which (78) makes possible, and split on the radius $r_S = \sqrt{2a\log S}$.

Distant centres, $m > r_S$. Here we use only $\rho_S \le \|\theta - b_{n,1}\|$, so that $\mathbb{E}[\rho_S^2 \mid \Delta Y_n] \le m^2 + K\sigma^2$. Since $\Delta Y_n \sim \mathcal{N}(\theta_0, I_K)$, the variable m^2 is noncentral chi-squared with K degrees of freedom and noncentrality $\|\theta - \theta_0\|^2$, so its tail obeys $\mathbb{E}[(m^2 + K\sigma^2)\mathbf{1}\{m > r\}] \le C(1 + r^K)e^{-(r - \|\theta-\theta_0\|)^2/2}$ for large r. At $r = r_S$ this is at most $C' S^{-a}(\log S)^{K/2} e^{\|\theta-\theta_0\|\sqrt{2a\log S}}$, so

$$S^{2/K}\,\mathbb{E}\big[\rho_S^2\,\mathbf{1}\{m > r_S\}\big] \le C' S^{2/K - a}(\log S)^{K/2} e^{\|\theta-\theta_0\|\sqrt{2a\log S}} \longrightarrow 0,$$

because $2/K - a < 0$ and the remaining factors are subpolynomial in S.

Nearby centres, $m \le r_S$. Put $T = \min\{\sigma^2/(2m), \sigma\}$. For $\|z - \theta\| \le t \le T$ we have $\|z - \Delta Y_n\|^2 \le m^2 + 2mt + t^2 \le m^2 + 2\sigma^2$, whence $f_n(z) \ge e^{-1} f_n(\theta)$ and therefore

$$F(t) \ge e^{-1}\omega_K f_n(\theta)\,t^K, \qquad 0 \le t \le T.$$

Substituting into (80) and extending the range of integration,

$$\int_0^{T^2} e^{-SF(\sqrt{v})}dv \le \int_0^\infty e^{-Se^{-1}\omega_K f_n(\theta) v^{K/2}}dv = \Gamma\big(1+\tfrac{2}{K}\big)\big(e^{-1}S\omega_K f_n(\theta)\big)^{-2/K}, \tag{81}$$

so that $S^{2/K}$ times the left side is at most $e^{2/K}\Gamma(1+\frac{2}{K})(\omega_K f_n(\theta))^{-2/K}$, a bound valid for every S and every centre.

For the remainder, $F(t) \geq F(T) \geq \beta := e^{-1}\omega_K f_n(\theta)T^K$ for $t \geq T$, and additionally $F(t) \geq \mathbb{P}(\sigma\chi_K \leq t - m)$ for $t \geq m$ because $B(\theta, t) \supseteq B(\Delta Y_n, t - m)$. Bounding the integrand by $e^{-S\beta}$ up to the radius at which the chi bound takes over, and by the chi bound beyond it,

$$\int_{T^2}^{\infty} e^{-SF(\sqrt{v})}dv \leq C(m+\sigma)^2 e^{-S\beta}.$$

On $\{m \leq r_S\}$ we have $f_n(\theta) \geq (2\pi\sigma^2)^{-K/2}S^{-a/\sigma^2}$ and $T \geq c/\sqrt{\log S}$ for large S, so $S\beta \geq c' S^{1-a/\sigma^2}(\log S)^{-K/2} \to \infty$ because $a < \sigma^2$. Hence $S^{2/K}(m+\sigma)^2 e^{-S\beta} \to 0$ uniformly over $\{m \leq r_S\}$.

Conclusion. On $\{m \leq r_S\}$ the quantity $S^{2/K}\mathbb{E}[\rho_S^2 \mid \Delta Y_n]$ is therefore bounded, for all large S, by

$$e^{2/K}\Gamma(1+\tfrac{2}{K})\big(\omega_K f_n(\theta)\big)^{-2/K} + 1,$$

and $\mathbb{E}[f_n(\theta)^{-2/K}] = (2\pi\sigma^2)\mathbb{E}[e^{m^2/(K\sigma^2)}]$ is finite precisely under (78), by the Gaussian moment identity at $a = 1/(K\sigma^2) < 1/2$. That is an integrable dominating function, and the indicator $\mathbf{1}\{m \leq r_S\}$ increases to one. Classical nearest-neighbour asymptotics give the pointwise limit (76) for each fixed ΔY_n, so dominated convergence yields (79); the distant-centre estimate shows the discarded region contributes nothing. Uniform integrability follows from the same dominating function. □

Retaining $\sigma^2 = 1 - h$ rather than passing to the limit, the constant in (77) reads

$$2\pi\sigma^2\,\Gamma(1+\tfrac{2}{K})\,\omega_K^{-2/K}\Big(\frac{K\sigma^2}{K\sigma^2-2}\Big)^{K/2}\exp\Big(\frac{\|\theta-\theta_0\|^2}{K\sigma^2-2}\Big), \tag{82}$$

which recovers (77) as $h \downarrow 0$. Condition (78) is $K(1-h) > 2$: it is the finite-h form of the same integrability boundary $K > 2$ that governs Corollary 3.7, and for $K \geq 3$ it holds for all small h.

The limit (82) has been checked numerically to five significant figures. Because the convergence is slow, direct simulation is the wrong instrument; but the quantity is a two-dimensional integral rather than an intrinsically stochastic object, since $\mathbb{P}(\|\theta - b\| \leq t \mid \Delta Y_n)$ is a noncentral chi-squared distribution function in closed form. Quadrature in the rescaled variable then reaches any S. At $K = 4$ the ratio of the computed value to (77) rises from 0.848 at $S = 10^3$ through 0.993 at $S = 10^8$ to 1.0000 at $S = 10^{18}$, and the same holds at $K = 3$, at $K = 6$, and at $\|\theta - \theta_0\| = 1.5$, always from below. Table 6 reports the computation.

Table 6: Convergence of $S^{2/K}\mathbb{E}[\min_s\|\theta - b_{n,s}\|^2]$ to the limit (82), computed by quadrature rather than simulation. Entries are the ratio of the computed value to the limit. Convergence is monotone from below and slow, which is why direct simulation at $S = 5{,}000$ reaches only about 89 per cent and settles nothing; the quadrature reaches $S = 10^{18}$ at negligible cost. The last column gives the limit itself.

K	$\|\theta-\theta_0\|$	10^3	10^5	10^8	10^{12}	10^{18}	limit
4	0	0.848	0.945	0.993	0.99974	0.999999	10.027
4	1.5	0.514	0.710	0.906	0.98732	0.999692	30.884
6	0	0.897	0.950	0.990	0.99935	0.999993	10.953
3	0	0.764	0.907	0.983	0.99875	0.999980	11.342

What Proposition A.3 settles is the first of the two requirements named above. It does not settle the second, and the following remark explains why that one is not a formality either.

The second requirement is a uniform-in-θ statement, and it follows from the same uniform integrability, together with the observation that the criterion is Lipschitz in θ with a constant that diverges only polynomially in S.

Proposition A.4 (Uniform law of large numbers)**.** *Let $\Theta \subset \mathbb{R}^K$ be compact with diameter D, assume* (78)*, and write*

$$X_N(\theta) = \frac{1}{N}\sum_{n=0}^{N-1} \psi_n(\theta), \qquad \psi_n(\theta) = S^{2/K} \min_{s \le S} \|\theta - b_{n,s}\|^2.$$

If

$$\frac{N}{\log S} \longrightarrow \infty, \tag{83}$$

then $\sup_{\theta\in\Theta}|X_N(\theta) - G(\theta)| \to 0$ *in probability, where G is the limit* (82)*.*

Proof. Write P_N for the empirical measure over n and P for the law of a single observation, so $X_N(\theta) = P_N\psi_\theta$.

Uniform integrability over Θ. The proof of Proposition A.3 bounds the conditional upper tail directly: for $u \le S^{2/K}T^2$,

$$\mathbb{P}(\psi_n(\theta) > u \mid \Delta Y_n) \le \exp(-e^{-1}\omega_K f_n(\theta) u^{K/2}),$$

a Weibull tail with scale $f_n(\theta)^{-2/K}$ and no dependence on S, while the region $u > S^{2/K}T^2$ is controlled by the two-region estimate of that proof. Consequently $\mathbb{E}[\psi_\theta \mathbf{1}\{\psi_\theta > \tau\}]$ is bounded by a quantity depending on ΔY_n alone which tends to zero as $\tau \to \infty$. Over $\theta \in \Theta$ the scale satisfies $f_n(\theta)^{-2/K} \le 2\pi\sigma^2 \exp\{(\|\Delta Y_n - \theta^*\| + D)^2/(K\sigma^2)\}$ for a fixed $\theta^* \in \Theta$, and since $(a + D)^2 \le (1+\eta)a^2 + (1+\eta^{-1})D^2$, this is integrable for η small enough exactly because (78) is strict. Hence

$$\lim_{\tau\to\infty} \sup_{S\ge 1} \sup_{\theta\in\Theta} \mathbb{E}\big[\psi_\theta \mathbf{1}\{\psi_\theta > \tau\}\big] = 0. \tag{84}$$

Entropy of the truncated class. Fix $\tau > 0$ and let $\mathcal{F}_\tau = \{\psi_\theta \wedge \tau : \theta \in \Theta\}$, which has constant envelope τ. Since $\theta \mapsto \min_s\|\theta - b_{n,s}\|$ is 1-Lipschitz and truncation is 1-Lipschitz,

$$\big|(\psi_\theta \wedge \tau) - (\psi_{\theta'} \wedge \tau)\big| \le L\,\|\theta - \theta'\|, \qquad L = 2S^{2/K} \sup_{\theta\in\Theta} \min_s \|\theta - b_{n,s}\|,$$

and $\|L\|_{P,2} \le 2S^{2/K}\{(\mathbb{E}\rho^2(\theta^*))^{1/2} + D\} \le CS^{2/K}$ for all large S, the moment being finite by Proposition A.3. A Lipschitz-in-parameter family over a K-dimensional set has bracketing numbers controlled by covering numbers of the parameter set, so

$$N_{[\,]}(\eta, \mathcal{F}_\tau, L^2(P)) \le \Big(\frac{CD\,S^{2/K}}{\eta}\Big)^K, \qquad \log N_{[\,]}(\eta) \le 2\log S + K\log(CD/\eta).$$

The bracketing entropy integral over the envelope is therefore

$$J_{[\,]}(\tau) = \int_0^\tau \sqrt{1 + \log N_{[\,]}(\eta)}\, d\eta \le C'\tau\big(\sqrt{\log S} + \sqrt{K}\big),$$

and the maximal inequality for bracketing entropy gives

$$\mathbb{E}\sup_{\theta\in\Theta}\big|P_N(\psi_\theta\wedge\tau) - P(\psi_\theta\wedge\tau)\big| \le \frac{C'\tau\big(\sqrt{\log S} + \sqrt{K}\big)}{\sqrt{N}}. \tag{85}$$

Combination. For every θ, $|P_N\psi_\theta - P\psi_\theta|$ is at most the left side of (85) plus $P_N[\psi_\theta\mathbf{1}\{\psi_\theta > \tau\}] + P[\psi_\theta\mathbf{1}\{\psi_\theta > \tau\}]$. The last two are small uniformly in θ by (84), the first of them by

Markov's inequality since its mean is the second. Choosing $\tau = \tau_N \to \infty$ slowly enough that $\tau_N \sqrt{\log S/N} \to 0$, which is possible precisely under (83), gives $\sup_\Theta |X_N - \mathbb{E}X_N| \to 0$ in probability.

From $\mathbb{E}X_N$ to G. By spherical symmetry $\mathbb{E}X_N(\theta)$ depends on θ only through $r = \|\theta - \theta_0\|$. It is nondecreasing in r: conditionally on ΔY_n the distribution function $t \mapsto \mathbb{P}(\|\theta - b_{n,1}\| \le t)$ is nonincreasing in $m = \|\theta - \Delta Y_n\|$, because a noncentral chi-squared distribution function is nonincreasing in its noncentrality, so $\mathbb{E}[\rho^2 \mid \Delta Y_n] = \int_0^\infty (1-F)^S$ is nondecreasing in m; and m^2 is stochastically nondecreasing in r. The limit G is continuous, so Proposition A.3 plus monotone pointwise convergence to a continuous limit on a compact interval gives $\sup_\Theta |\mathbb{E}X_N - G| \to 0$ by Pólya's argument. Combining the two displays completes the proof. $\square$

The condition (83) is weak. It permits S to grow at any polynomial or even sub-exponential rate in N, and in particular it is compatible with the published rate conditions (60), which constrain N from above through $N/S^{1/4} \to 0$: the pair $N \asymp S^{1/8}$ satisfies both.

The distinction between the two routes is visible numerically, and Table 7 was computed to settle it before the proof above was written. The discriminating statistic is the ratio of the uniform deviation to the pointwise deviation, which cancels both the $N^{-1/2}$ and the scale of the summand: it grows like $S^{1/K}$ if the criterion's divergent Lipschitz constant governs the uniform behaviour, and like $\sqrt{(2/K)\log S}$ if the envelope does. Across two decades of S the observed ratio grows by a factor of 1.35, against 1.41 for the logarithmic prediction and 2.51 for the power. The individual values track the logarithm to within a few per cent and are nowhere near the power.

Table 7: Uniform against pointwise deviation of the nearest-atom criterion, at $K = 5$ with $N = 150$ observations, 10 replications and 21 values of θ spanning $[0, 1.5]$. Deviations are taken about the pooled mean across replications, which shrinks both columns by the same factor and so leaves the ratio unaffected. The last two columns are the predictions of the two chaining routes, normalised to nothing: what matters is their growth. The ratio follows the logarithm.

S	sup dev.	pointwise dev.	ratio	$S^{1/K}$	$\sqrt{(2/K)\log S}$
10^2	0.970	0.690	1.407	2.512	1.357
10^3	1.280	0.674	1.899	3.981	1.662
10^4	1.082	0.569	1.902	6.310	1.919
growth			$\times 1.35$	$\times 2.51$	$\times 1.41$

Remark A.5 (A rate condition we previously reported, in error)**.** An earlier version of this appendix asserted that the uniform statement plausibly required $N \gg S^{2/K}$, on the ground that each summand has Lipschitz constant of order $S^{2/K}\rho \asymp S^{1/K}$, which diverges. The Lipschitz constant does diverge, and the criterion is genuinely rough at scale $S^{-1/K}$; the error was in what that implies. Chaining was applied over the metric diameter of the index set, which is of order $S^{1/K}$, instead of over the envelope of the function class. After truncation the envelope is bounded, the diverging Lipschitz constant enters the entropy only through its logarithm, and (85) costs $\sqrt{\log S}$ rather than $S^{1/K}$. The correct condition is (83), and the direction of the earlier claim, that the requirement pointed the opposite way to the published conditions, was wrong: there is no tension.

With both requirements discharged, the argmax question in this model can be settled in one direction.

Corollary A.6 (The simulated maximiser is consistent in the location model)**.** *Assume* (78) *and* (83)*, and suppose in addition*

$$\frac{M}{S^{2/K}\log S}\longrightarrow\infty. \tag{86}$$

Then the simulated maximum likelihood estimator in the model (73) *satisfies* $\widetilde{\theta}_{N,M,S}\to\theta_0$ *in probability. Along* $M=S=n$ *with* $K>2$*, condition* (86) *holds automatically, so consistency requires only* $N/\log n\to\infty$ *— along the very sequence for which Corollary 3.7 makes the simulated transition density converge in probability to zero.*

Proof. Multiply (75) by $S^{2/K}/N$ and set

$$A_N(\theta)=-\frac{2hS^{2/K}}{N}\log\widehat{\mathcal{L}}_{N,M,S}(\theta)-S^{2/K}hK\log(2\pi h),$$

so that $\sup_\theta|A_N(\theta)-X_N(\theta)|\le 2S^{2/K}h\log S\to 0$ by (86). The subtracted term does not depend on θ, so $\widetilde{\theta}_{N,M,S}$ minimises A_N. By Proposition A.4, $\sup_\Theta|A_N-G|\to 0$ in probability. The limit $G(\theta)=C\exp\{\|\theta-\theta_0\|^2/(K\sigma^2-2)\}$ is continuous and strictly increasing in $\|\theta-\theta_0\|$, hence uniquely minimised at θ_0 and well separated: $\inf_{\|\theta-\theta_0\|\ge\varepsilon}G(\theta)>G(\theta_0)$ for every $\varepsilon>0$. The argmin theorem for uniformly convergent criteria with a well-separated minimum gives the conclusion. For $M=S=n$, (86) reads $n^{1-2/K}\gg\log n$, which holds for every $K>2$. □

This is worth stating plainly, because it cuts against a reading the density results invite. Along $M=S=n$ with $K>2$ the simulated transition density collapses to zero at every fixed pair of points, and yet the maximiser of the simulated likelihood converges to the true parameter. In this model the density counterexample does not transfer, and Corollary A.6 shows it cannot be made to transfer by any argument that proceeds through pointwise density behaviour alone. The caution stated throughout this paper — that Corollary 3.7 refutes two published statements about the transition density without showing the estimator inconsistent — is not a hedge awaiting resolution; in the most tractable case it is provably the right conclusion.

What remains open is narrower and is precisely the $\sqrt{N}$ statement. Consistency is settled by Corollary A.6; what is not settled is the *rate*, and there the limit criterion disagrees with the exact one. Its curvature at θ_0 differs from the exact maximum-likelihood curvature. Writing the limit (77) as $C_K\exp\{\|\theta-\theta_0\|^2/(K-2)\}$, its Hessian at θ_0 is $2C_K/(K-2)$ times the identity, so the comparison carries the constant C_K as well as the factor $2/(K-2)$; the exact criterion has Hessian $2I$. The fluctuation of $N^{-1}S^{2/K}D_N$ about its limit is $O_{\mathbb{P}}(N^{-1/2})$ with a covariance not depending on S. Then $\sqrt{N}(\widetilde{\theta}_{N,M,S}-\widehat{\theta}_N)$ would converge to a nondegenerate limit rather than to zero, for every admissible (M_n,S_n) including the regime in which Theorem 2 of Brandt and Santa-Clara (2002) operates. That would refute the *conclusion* of Lemma 5, not merely its proof, and would change the standing of this paper's main claim.

Three things stand between Proposition A.4 and that conclusion, and none is supplied here. First, a rate for $\mathbb{E}X_N-G$: Proposition A.3 gives convergence but not speed, and the quadrature of Table 6 shows it is slow, so the curvature comparison needs $\mathbb{E}X_N-G=o(N^{-1})$ uniformly near θ_0 and that is not established. Second, a functional central limit theorem for $\sqrt{N}(X_N-\mathbb{E}X_N)$ near the minimiser, where the summands are not differentiable in θ and, for $K\sigma^2\le 4$, not square integrable either. Third, the sandwich slack must be negligible against the *variation* rather than the level, which by Remark A.7 means $M\gg NS^{2/K}\log S$ rather than (86). The second of these is the substantive one.

Remark A.7 (Why Proposition A.2 cannot deliver this)**.** The uniform sandwich has slack $2h\log S$ after division by N. Against the *level* $D_N/N\asymp S^{-2/K}$ that slack is negligible whenever

$M \gg S^{2/K} \log S$, which along $M = S$ with $K > 2$ holds comfortably. So Proposition A.2 is strong enough to transfer consistency, and Corollary A.6 does exactly that now the limit is available. It is not strong enough for the $\sqrt{N}$ statement, because there the comparison is with the *variation* of D_N/N over a ball of radius $N^{-1/2}$ around the minimiser, where the limit criterion is flat and the slack is not negligible. Any route to the $\sqrt{N}$ conclusion must therefore go through the limit (77) rather than through the sandwich.

Remark A.8 (An existing literature that may supply the machinery)**.** The estimating equation just described is a *nonparametric simulated maximum likelihood* problem, and that object has been studied on its own terms. Kristensen and Shin (2012) analyse simulated likelihood in which the transition density is replaced by a kernel estimate built from simulated draws, treating the bandwidth and the simulation size as joint design parameters, and give conditions under which the resulting estimator inherits the first-order asymptotics of the infeasible exact-likelihood estimator. That is the same construction as Proposition A.1, and their conditions are the natural place to look for the uniform score control that hypothesis (iii) needs.

One structural difference decides whether their results transfer, and we flag it rather than assert an answer, not having attempted the transfer. In their setting the bandwidth is a free design parameter, to be chosen by the analyst and in particular to be sent to zero as slowly as the theory requires. In the endpoint estimator it is not free: it is $\sqrt{h} = M^{-1/2}$, pinned to the discretisation step by the construction itself, and the same M that sets the bandwidth also sets the Euler bias. Whether their conditions can be met under that constraint is exactly the question Corollary 6.8 bears on, since a bandwidth tied to M cannot be widened to buy variance without worsening bias. If their conditions can be met, Section A shortens considerably; if they cannot, and cannot for the reason just given, that is itself a sharper statement of the problem than the present appendix contains. Either outcome is worth having, and we record the comparison as the most promising unexplored route.

Remark A.9 (A likely shorter route)**.** Proposition A.1 makes the simulated transition density an exact kernel density estimate, so the simulated score is $\nabla_\theta \log \widehat{f}_n(\theta)$ and the estimating equation is $\sum_n \nabla_\theta \log \widehat{f}_n(\theta) = 0$. The asymptotics of log-density-gradient estimators at bandwidth $\sqrt{h}$ from S draws are standard, and would deliver directly the order of the discrepancy between the simulated and exact scores. That is exactly what hypothesis (iii) of Proposition 6.2 requires, and it is what Lemma 4 of Brandt and Santa-Clara (2002) asserts without proof. This route stays inside Section 6 rather than relying on nearest-neighbour theory, and we expect it to be both easier and more general. We record it as the first thing to try.

The status of the ingredients above should be kept distinct.

Proposition A.1 is an exact identity. Proposition A.2 is an exact uniform bound, and nothing more: it compares two criteria, it does not compare their maximisers. The limit of D_N is open, and with it the argmax question.

The nearest-neighbour limit is a *conjecture*. It is stated as such in the accompanying notes, it is used nowhere in this paper, and we do not assert that $N^{-1}S^{2/K}D_N(\theta)$ converges to the shape given there. A proof would require uniform-in-θ nearest-neighbour asymptotics for a triangular array whose atoms are exchangeable rather than independent, since they share the observation innovation, together with control of the dependence that common random numbers induce across θ and an argmax-continuity argument. None of these is available to us.

The accompanying notes also report *numerical observations*. Across the designs examined the simulated maximiser appears to approach θ_0, while the scaled gap $\sqrt{N}\|\widehat{\theta}_{N,M,S} - \widehat{\theta}_N\|$ shows no downward trend. Those runs use a single seed per design, cover one dimension, and cannot reach the regime $N^4 \ll S \ll M^2$ in which the published theorem operates. They are not a demonstration of consistency, and they are deliberately confined to the notes rather than reported as a finding here.

Accordingly the manuscript claims only what Section 7 establishes: Lemmas 2 and 3 are directly refuted, and the proofs of Theorems 1 and 2 therefore fail.

B Reproducibility

The accompanying directory contains:

- `main.tex`, the manuscript source;
- `references.bib`, the bibliography;
- `code/generate_results.py`, the fully reproducible simulation and figure code;
- vector PDF and PNG versions of all figures;
- CSV and LaTeX versions of generated numerical tables.

The code uses a fixed seed, vectorised Gaussian simulation, and log-sum-exp evaluation of the endpoint estimator. No proprietary data are used.

Environment and cost. The results in this paper were produced with Python 3.13.5, NumPy 2.3.5, SciPy 1.15.3, pandas 2.3.3 and Matplotlib 3.10.8, from the single seed 20260726 set in `code/generate_results.py`. The full artefact regeneration, which includes the counterexample path of Table 5 and the six-design grid sweep reported in the companion note, takes a few minutes on one desktop core; no run requires special hardware, and none is long enough to warrant checkpointing.

Verification. Three checks are automated and can be run by a reader. `make verify` regenerates every figure and table into a scratch directory and compares them with the committed copies, field-wise at relative tolerance 10^{-9} for CSV output and by SHA-256 for PDF and PNG; figure output carries no embedded timestamp, so the comparison is exact rather than approximate. `make test` runs the analytical test suite, in which every closed form quoted here is checked against an independent computation. `code/verify_dois.py` checks each bibliography DOI against the Crossref registry by title.

Archive. The code and manuscript are deposited, not merely described. The concept DOI `10.5281/zenodo.21719868` resolves to the most recent version and is the one to cite; each release additionally receives its own version DOI, and superseded versions remain resolvable and are listed with the specific correction that supersedes them.

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